

SHETH LUJ & MV COLLEGE

PRACTICAL NO 4

AIM : Performing Data Analysis using Calculations in Tableau

- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

A: Creating Basic Calculated Fields Using Formulas

1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your store_sales_data.csv, and click Open.

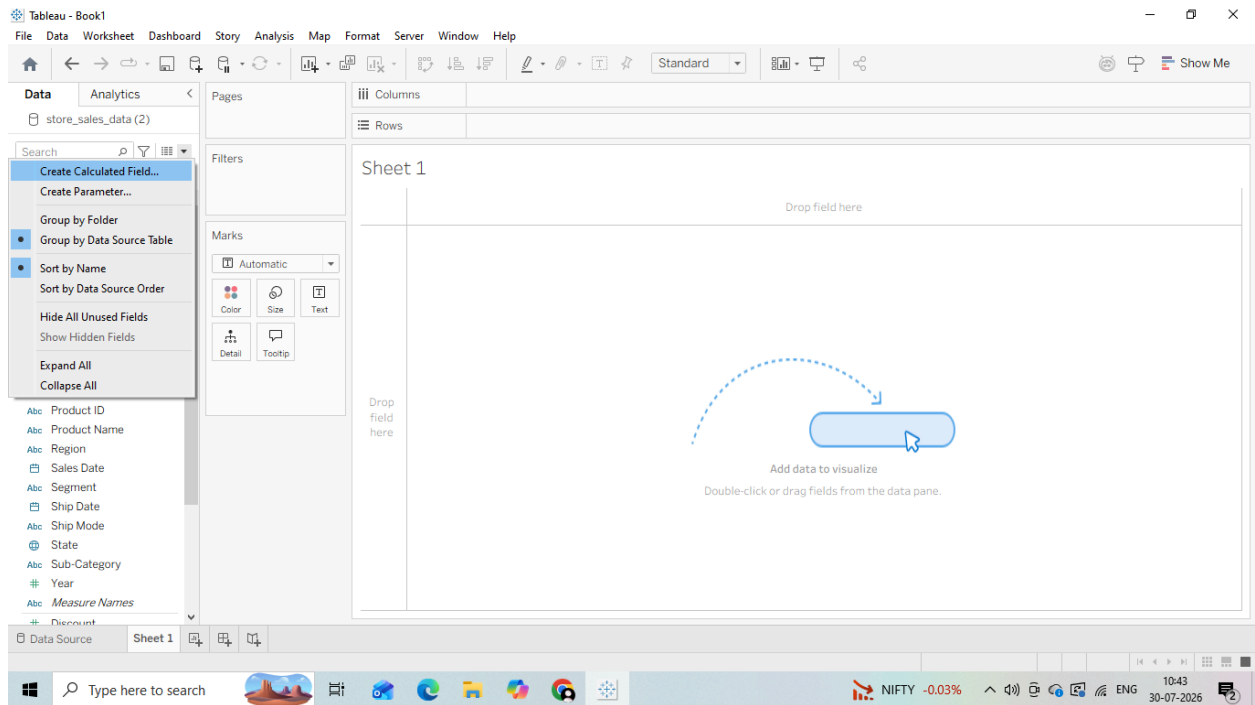
The screenshot shows the Tableau Desktop interface. On the left sidebar, under 'Connections', the file 'store_sales_data (2)' is selected. Below it, under 'Files', the same file is listed. The main workspace displays the data source 'store_sales_data (2).csv' with a preview of 5 fields: Customer ID, Customer Name, Last Name, Date of Birth, and Sales. The preview table shows 5 rows of data. The bottom status bar indicates 'Data Source' and 'Sheet 1'.

Type	Field Name	Physical Table	Rem...
Abc	Customer ID	store_sales_data...	Custo...
Abc	Customer Name	store_sales_data...	Custo...

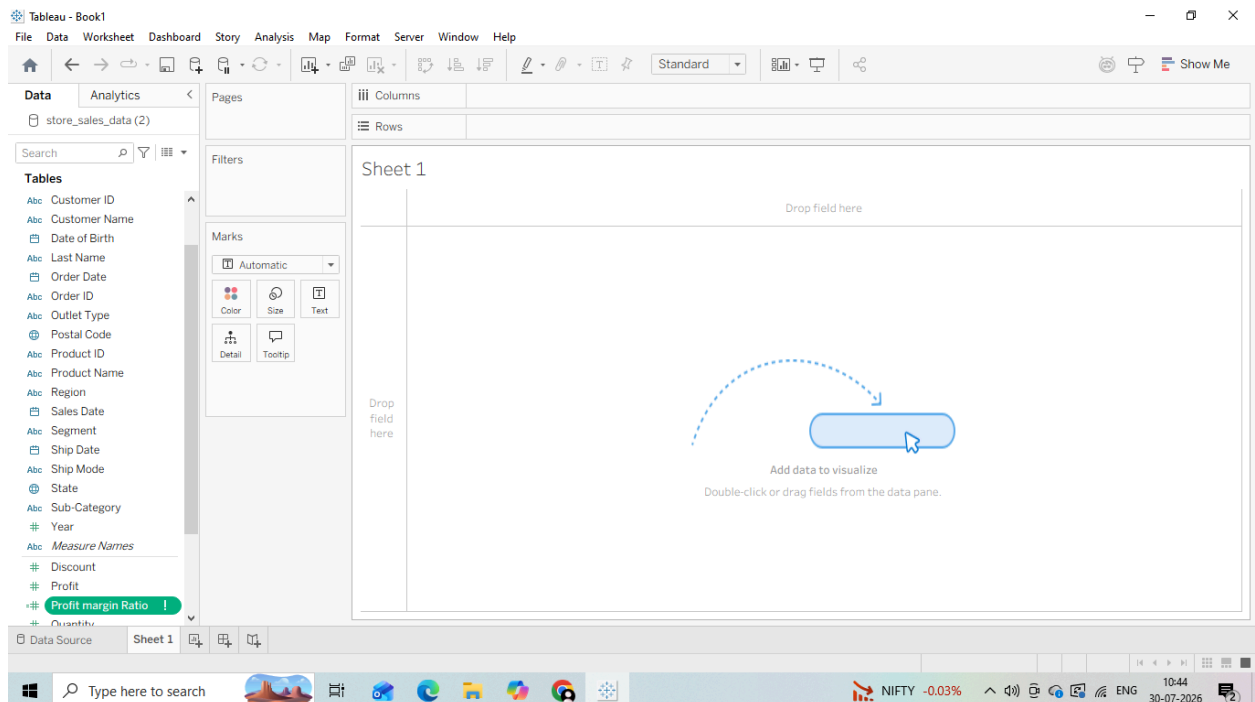
Abc	Abc	Abc	store_sales_data (2).csv	#
store_sales_data (2).csv	store_sales_data (2).csv	store_sales_data (2).csv	store_sales_data (2).csv	store_sales_data (2).csv
Customer ID	Customer Name	Last Name	Date of Birth	Sales
CUST000001	Curtis	Krause	29-11-1973	3.34
CUST000002	Jeremy	Foster	30-08-1987	37.05
CUST000003	Ariel	Mccann	15-12-1980	35.55
CUST000004	Colleen	Maynard	23-05-1989	8.96
CUST000005	Gina	Holden	31-12-1957	35.53

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2. Building a Basic Custom Calculation

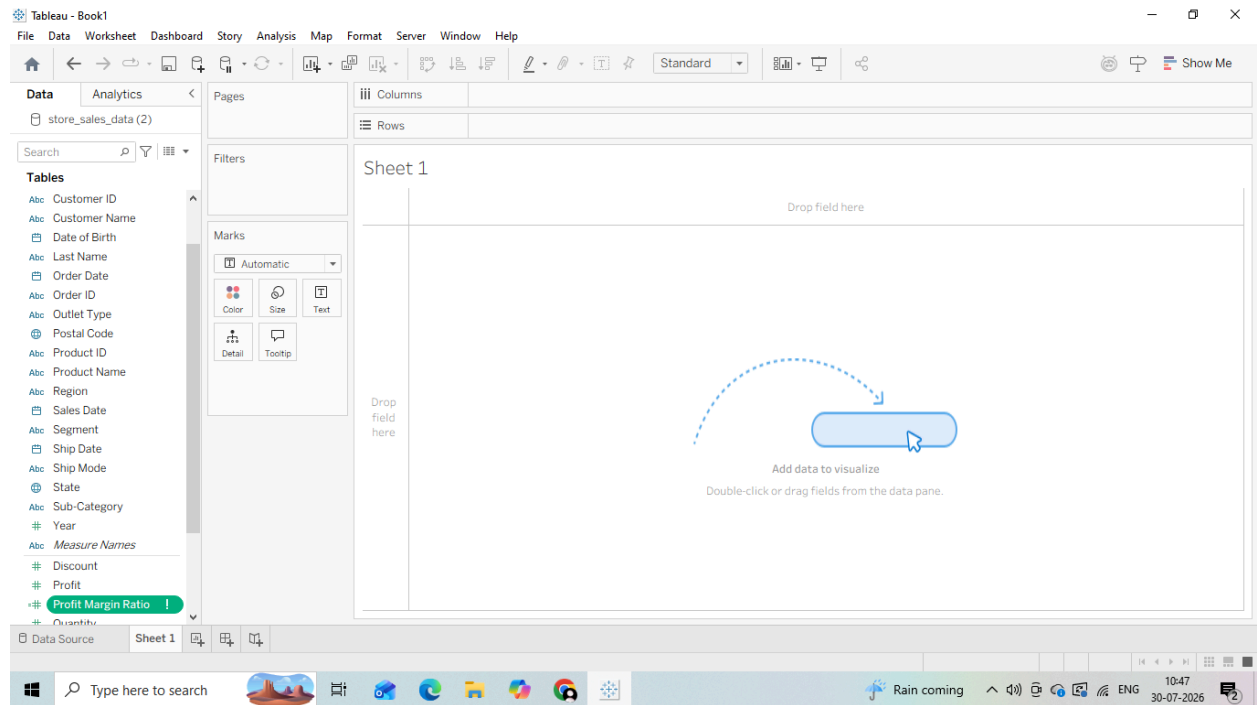


2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.

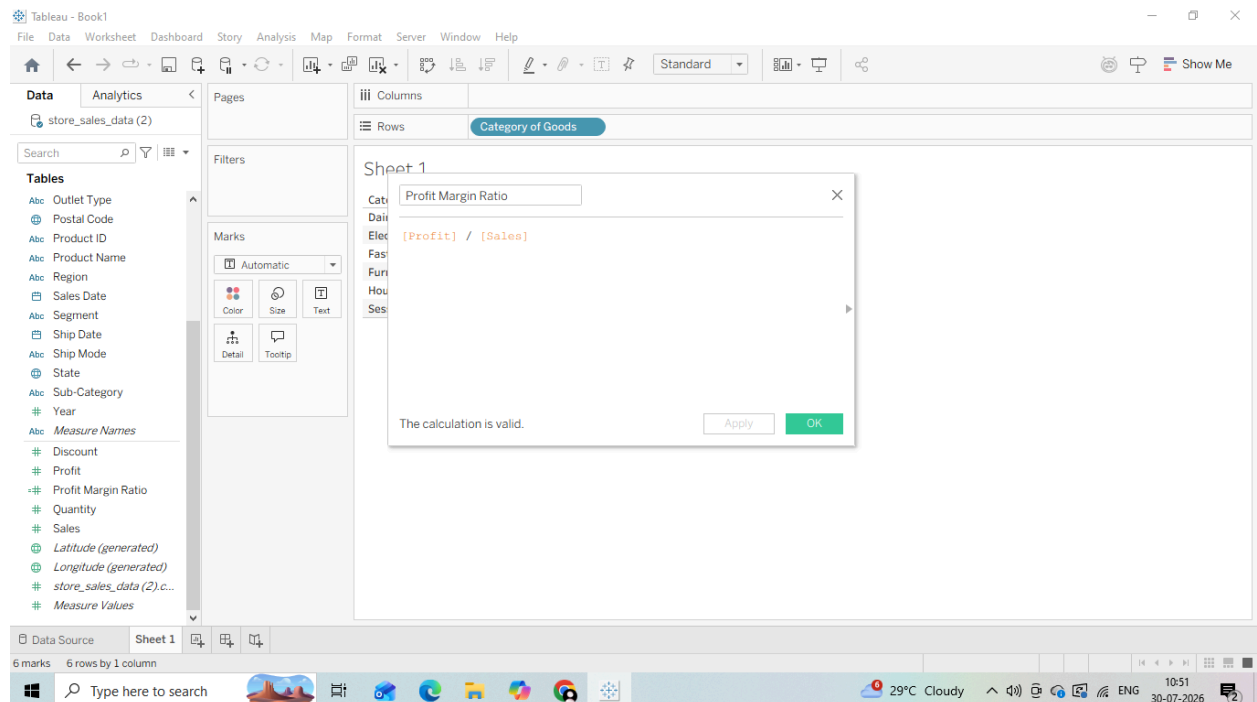


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3. In the main workspace box, enter the following basic mathematical formula:

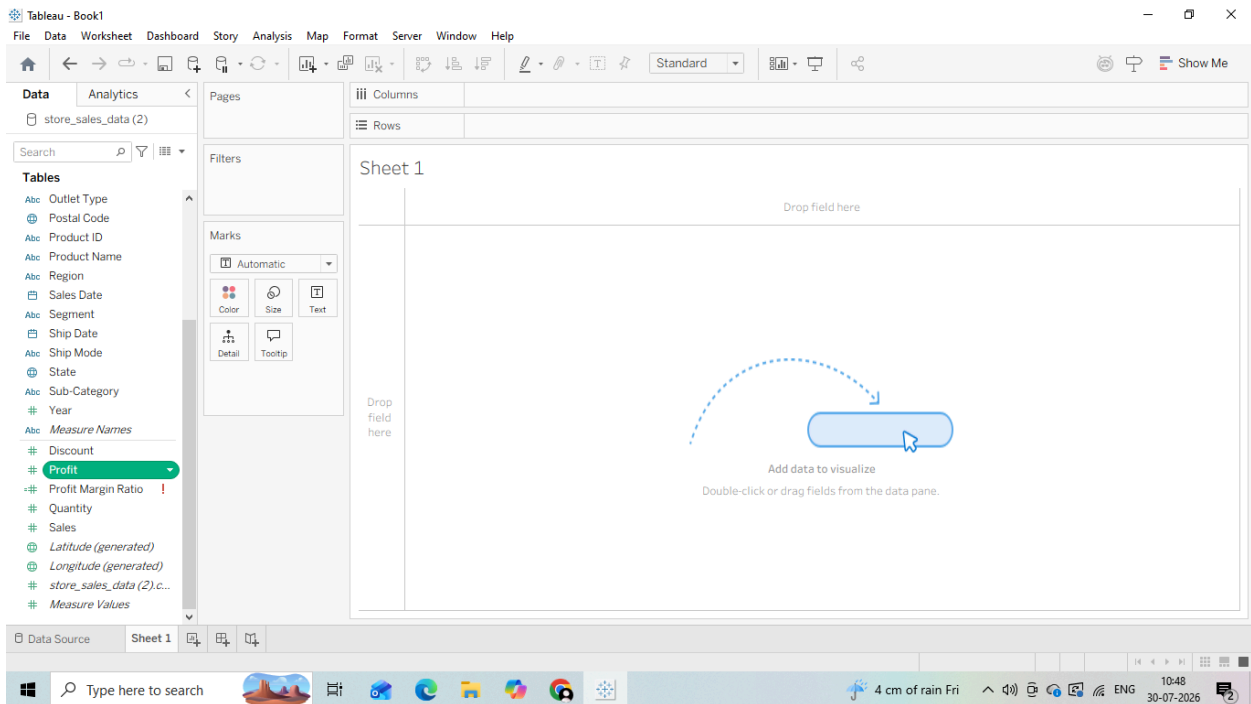


4. Look at the bottom-left corner of the editor box. It should say "The calculation is valid." Click OK.

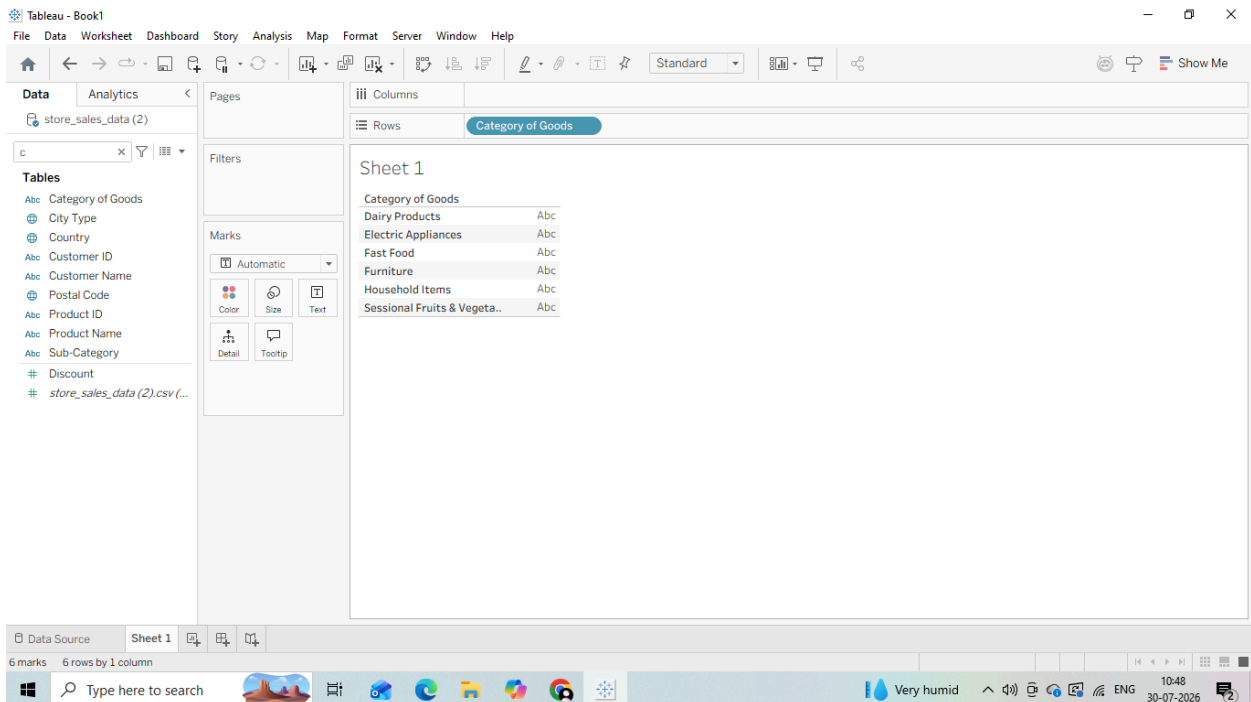


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5. Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.

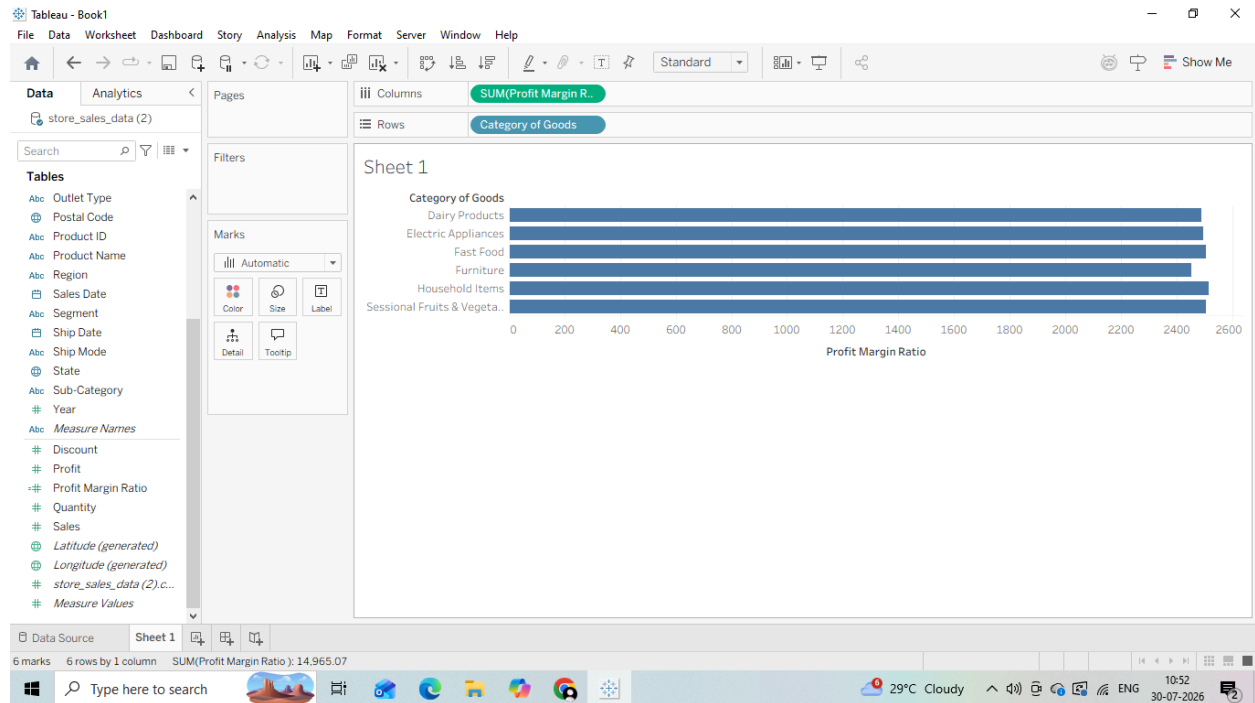


2. Putting the Field to Use



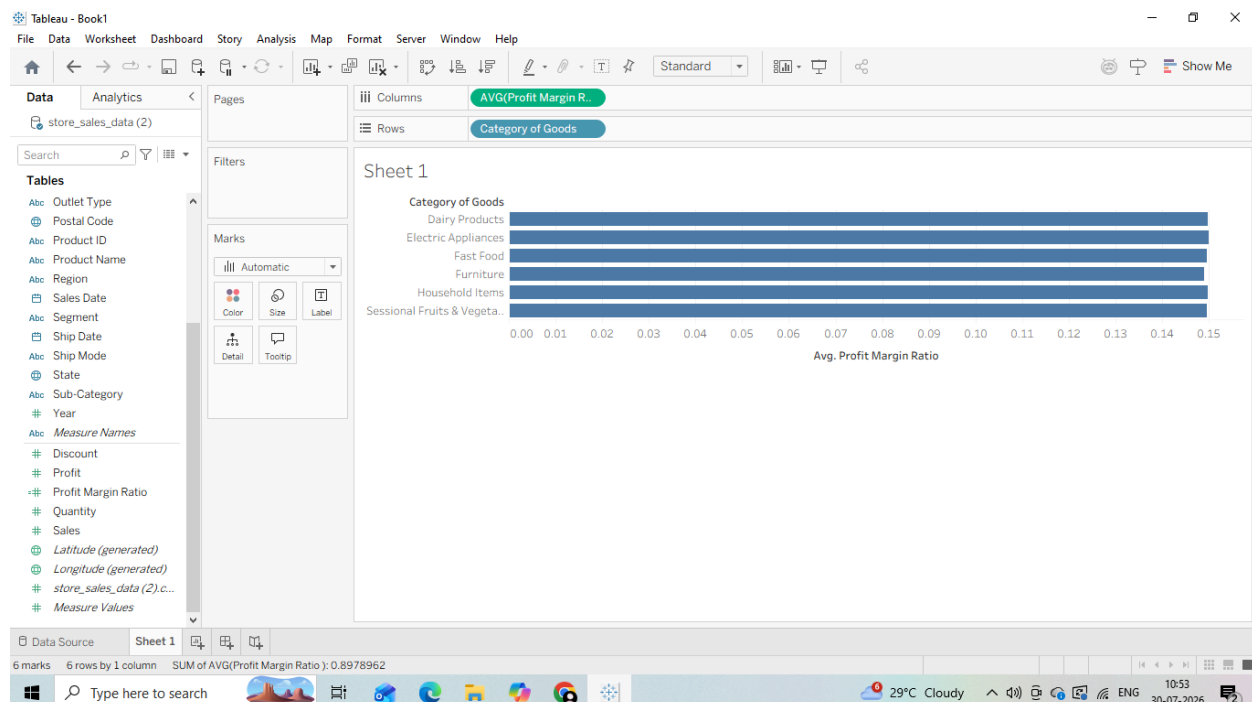
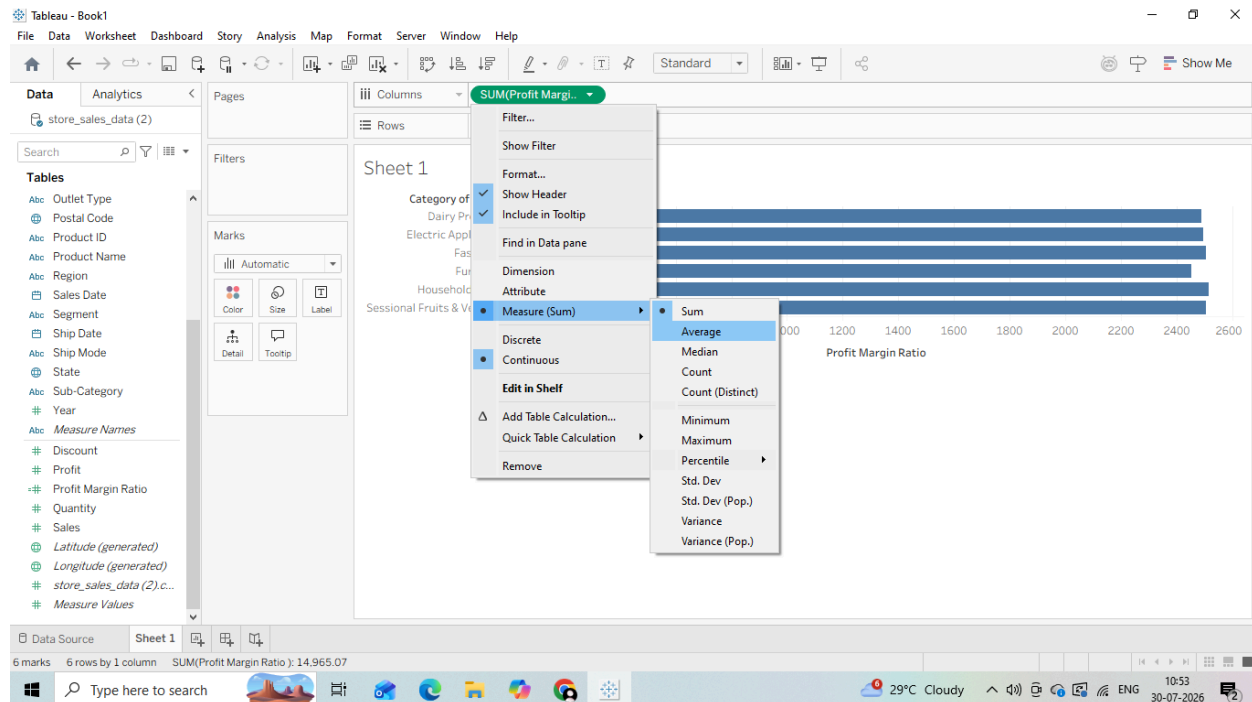
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2. Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.



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3. Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.

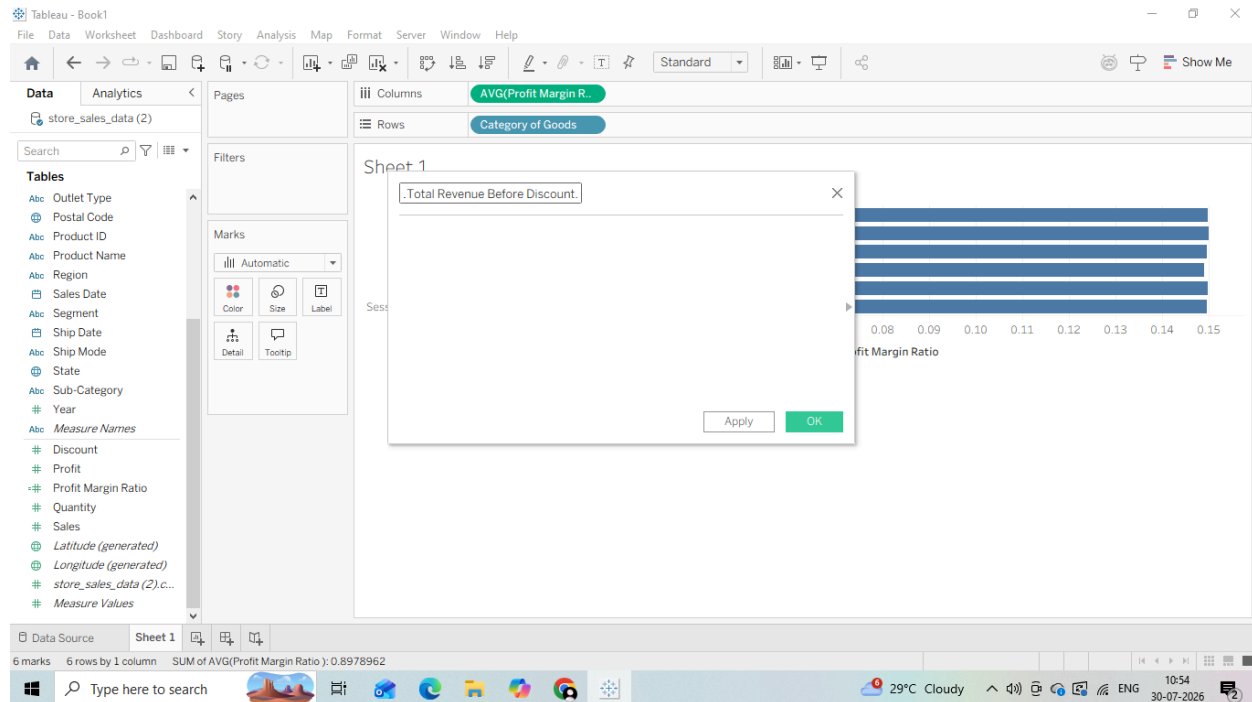


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B: Performing Row-Level Calculations

1. Creating a Row-Level Calculated Field

1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...

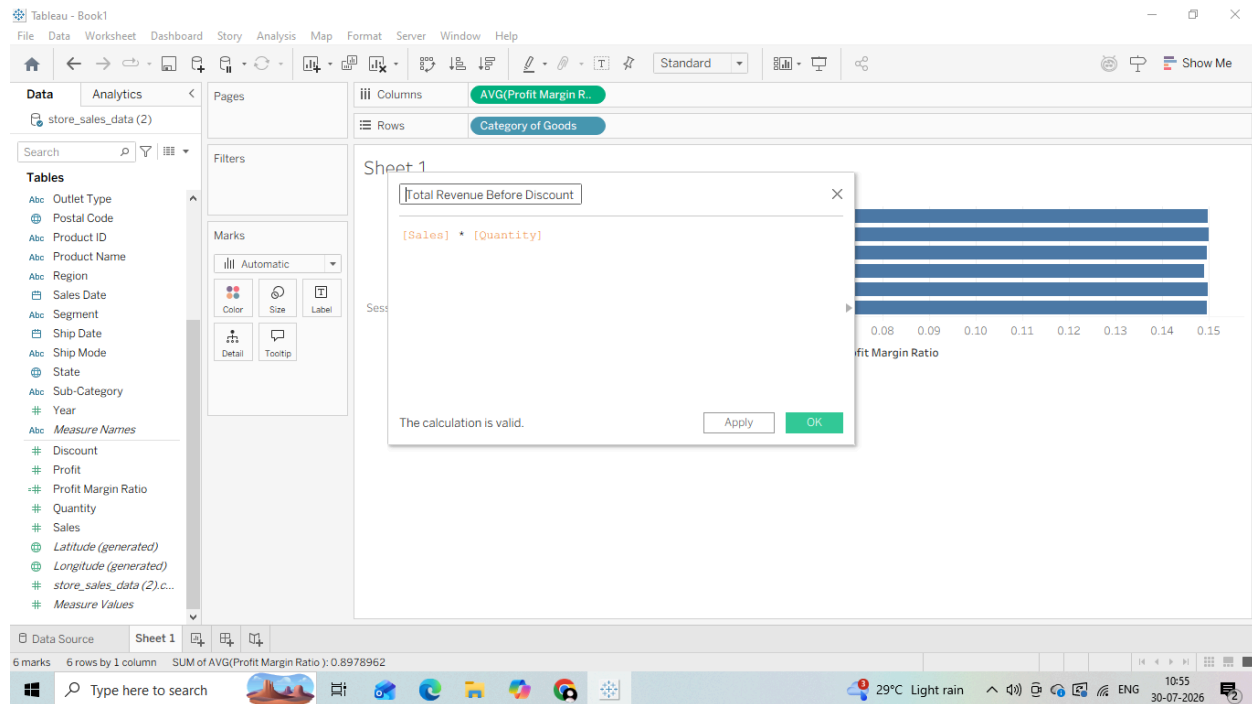


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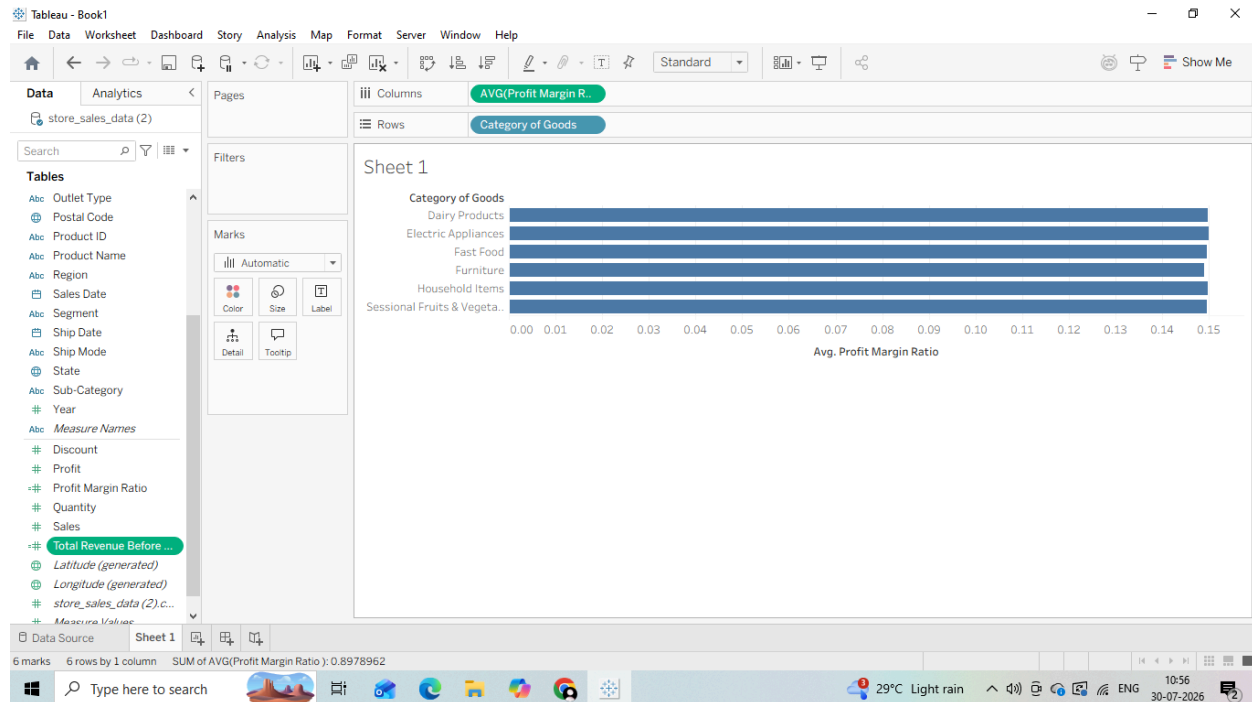
2. Name the field: Total Revenue Before Discount.

3. In the calculation formula canvas area, input the following expression:

$[Sales] * [Quantity]$



4. Verify the message at the bottom reads "The calculation is valid" and click OK.



2. Putting the Field to Use

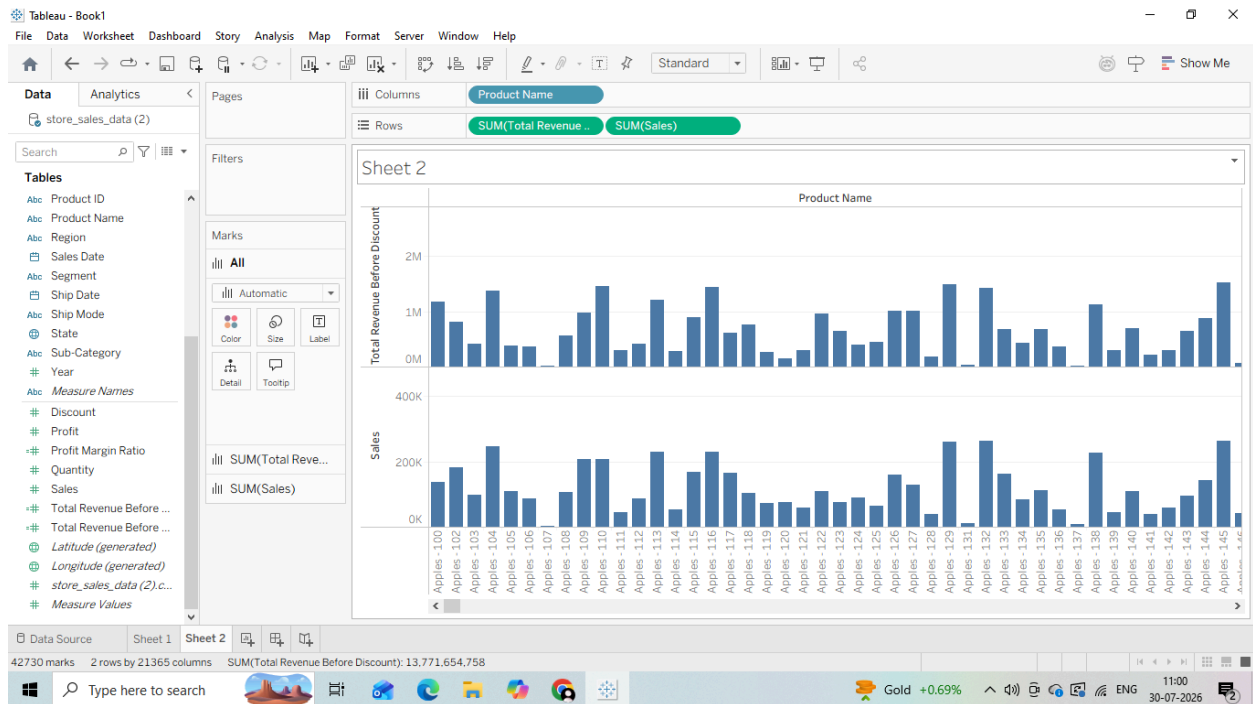
1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.

2. Drag your newly created Total Revenue Before Discount calculated field and drop

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it onto the Columns shelf.

3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.

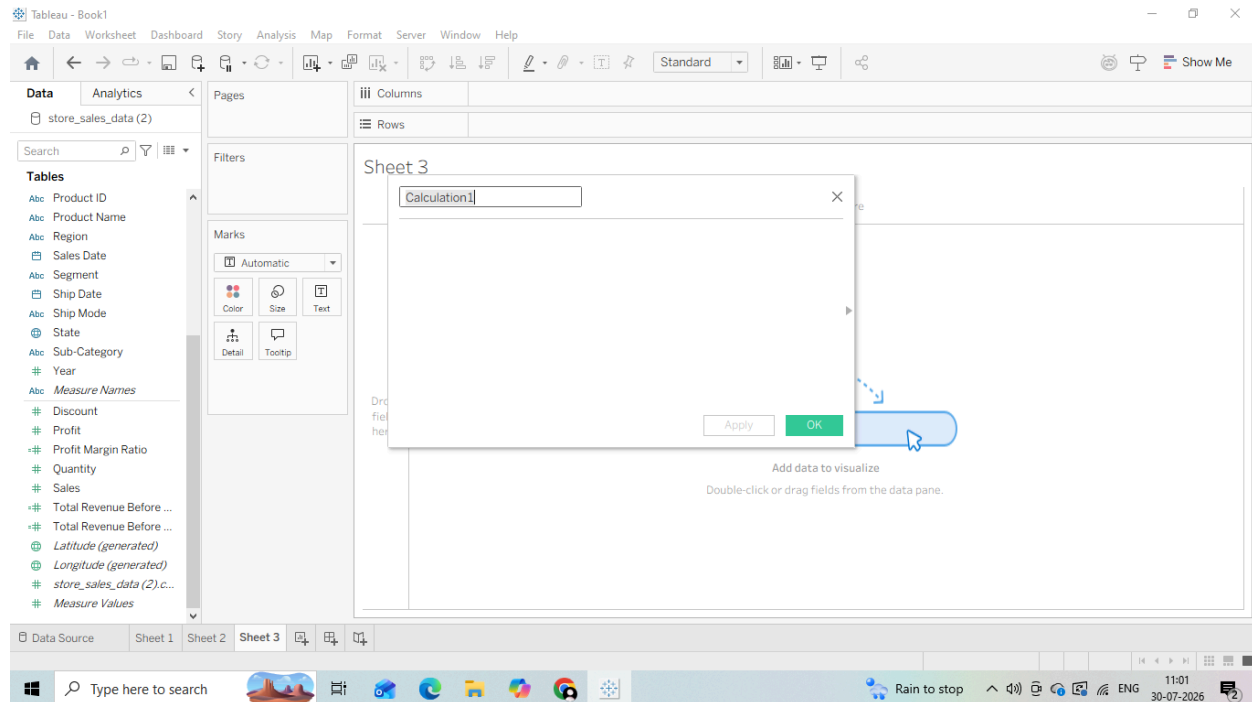


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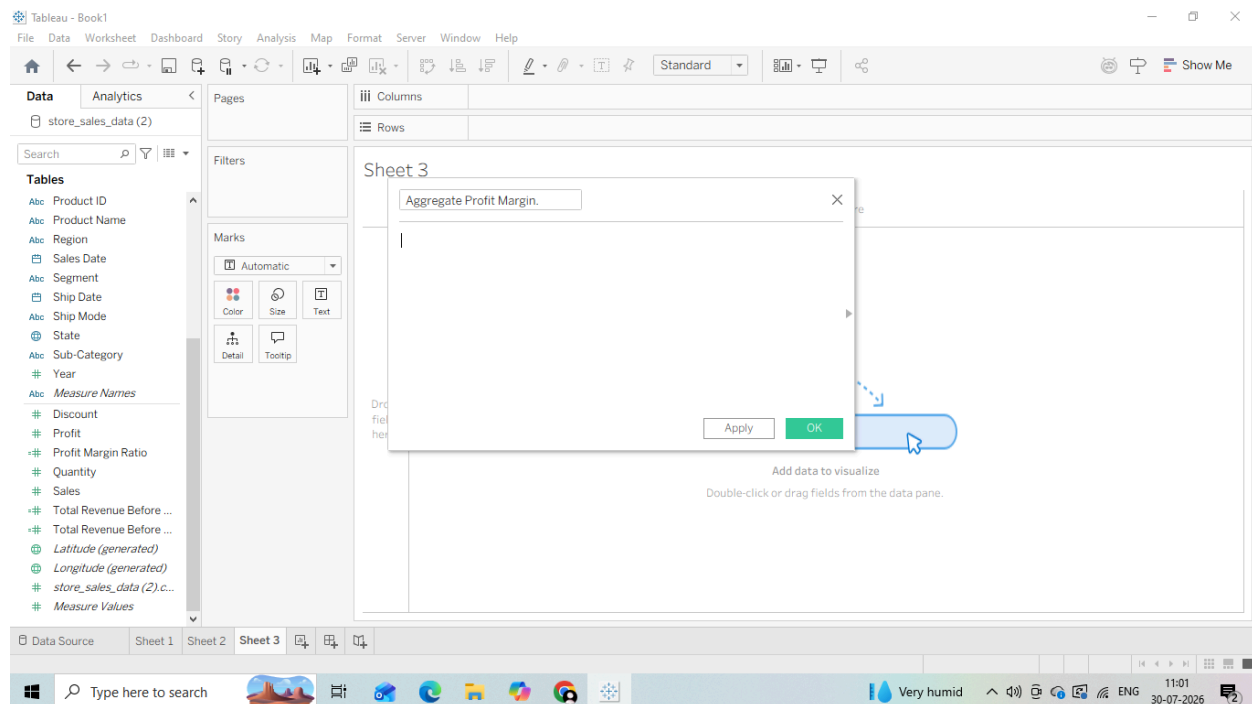
C: Performing Aggregate-Level Calculations

1. Creating an Aggregate Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...

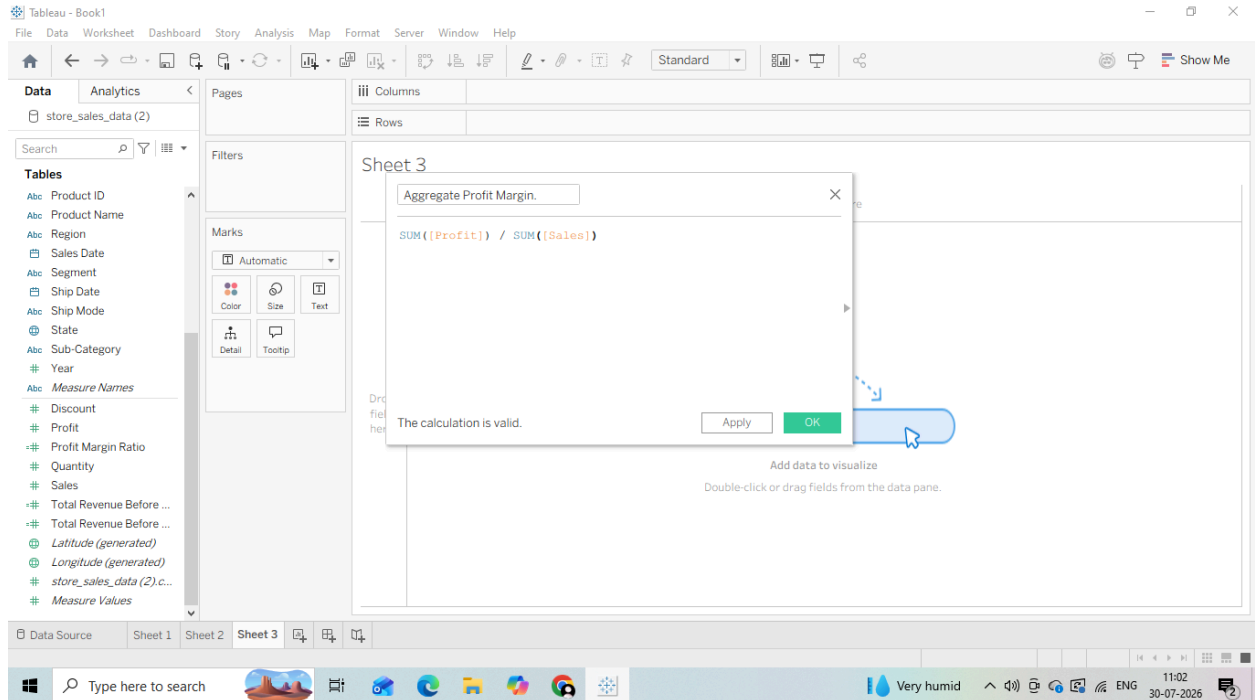


2. Name the field: Aggregate Profit Margin.

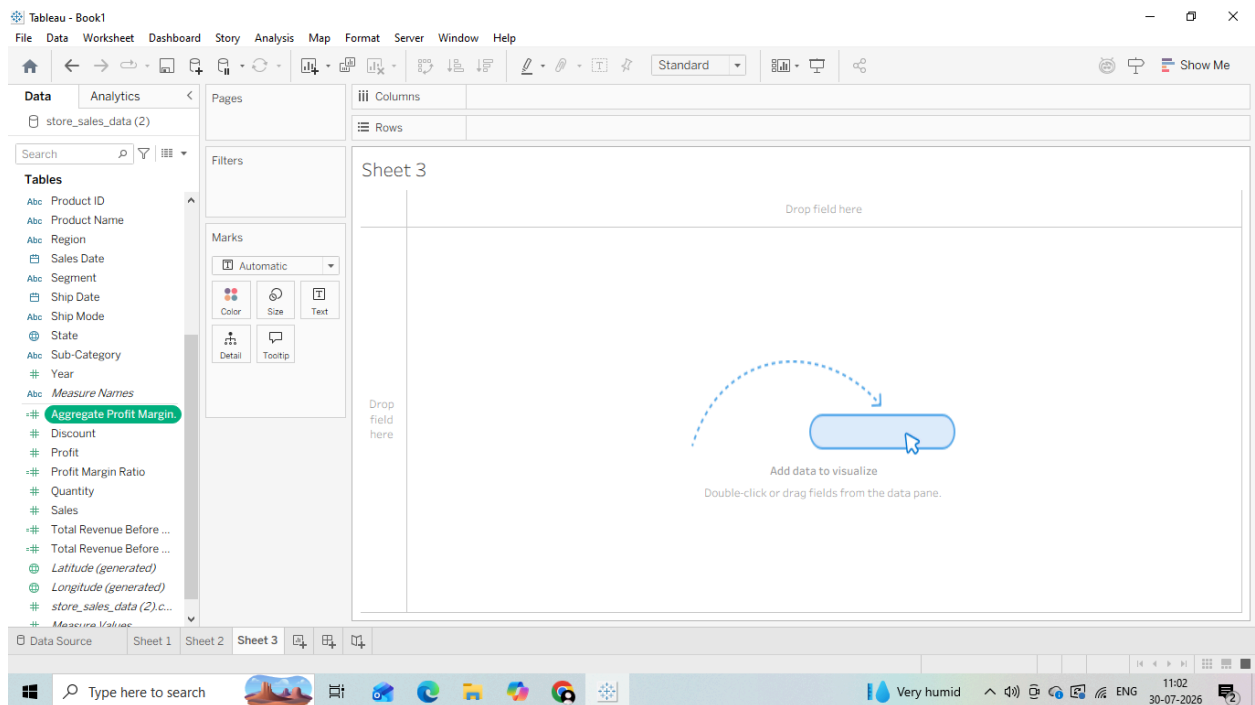


3. In the calculation formula window, type the following expression:
$$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$$

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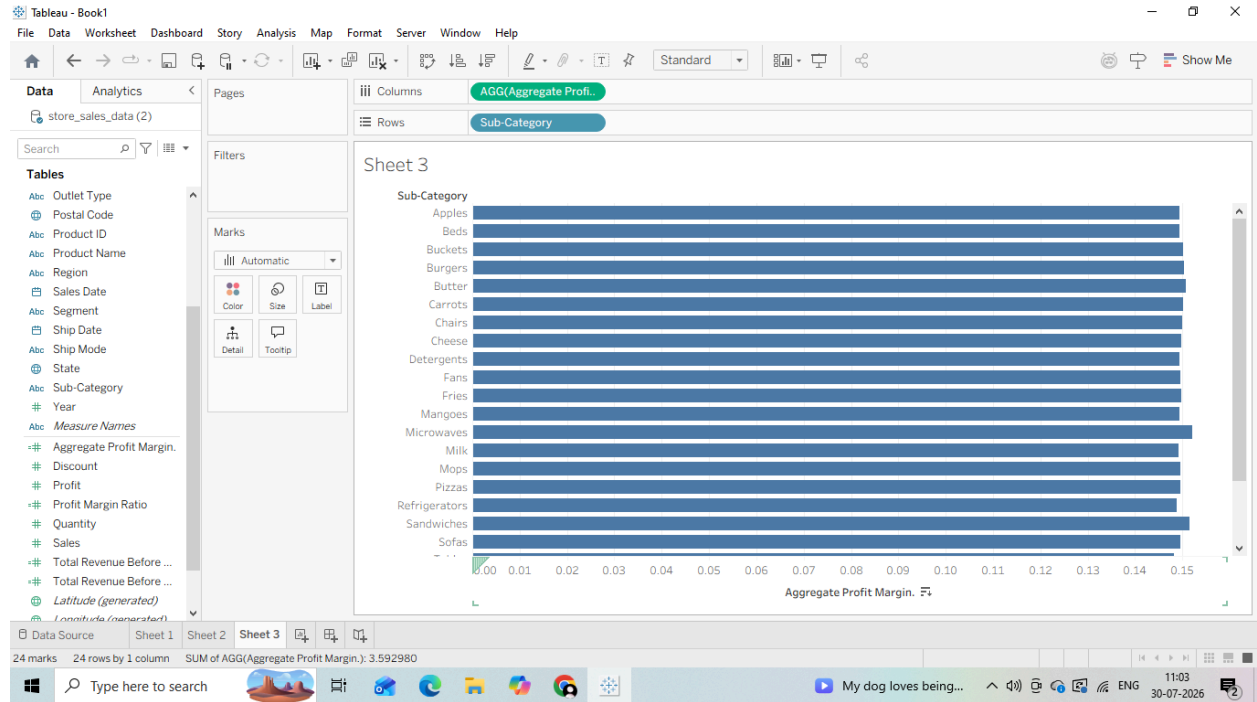
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



2. Putting the Field to Use

1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.

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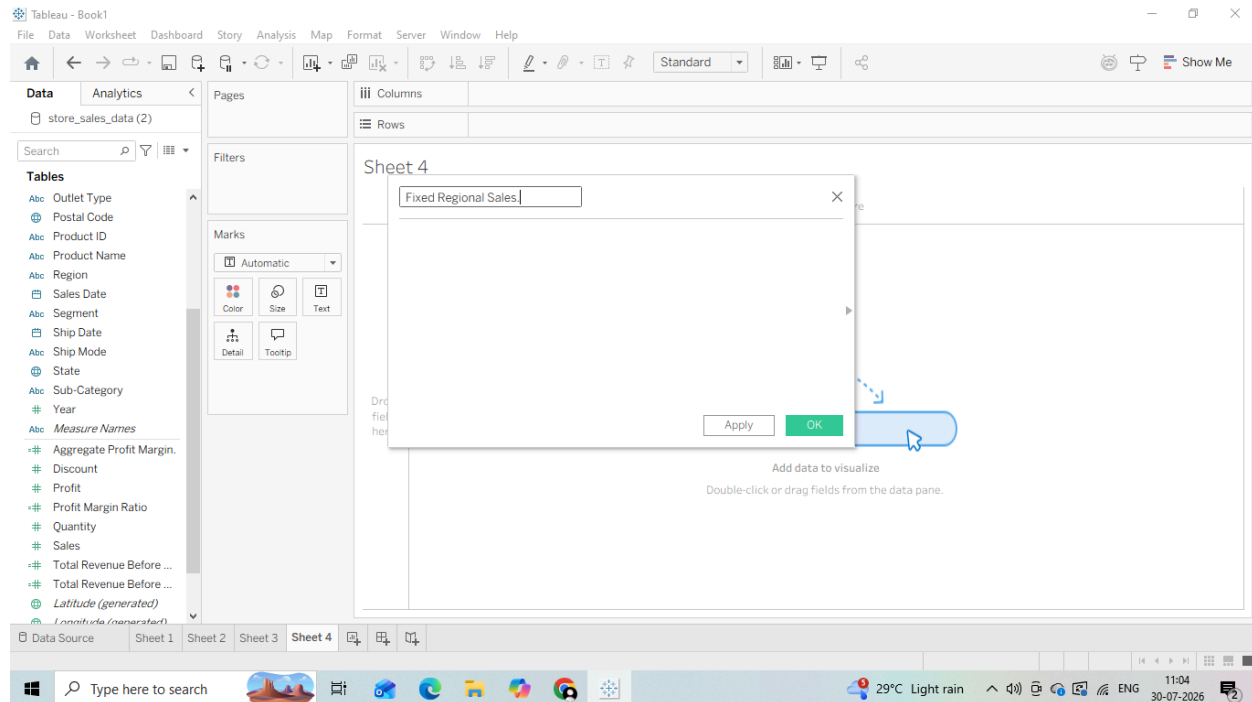


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D: Implementing FIXED Level of Detail (LOD) Calculations

1. Creating a FIXED LOD Calculated Field

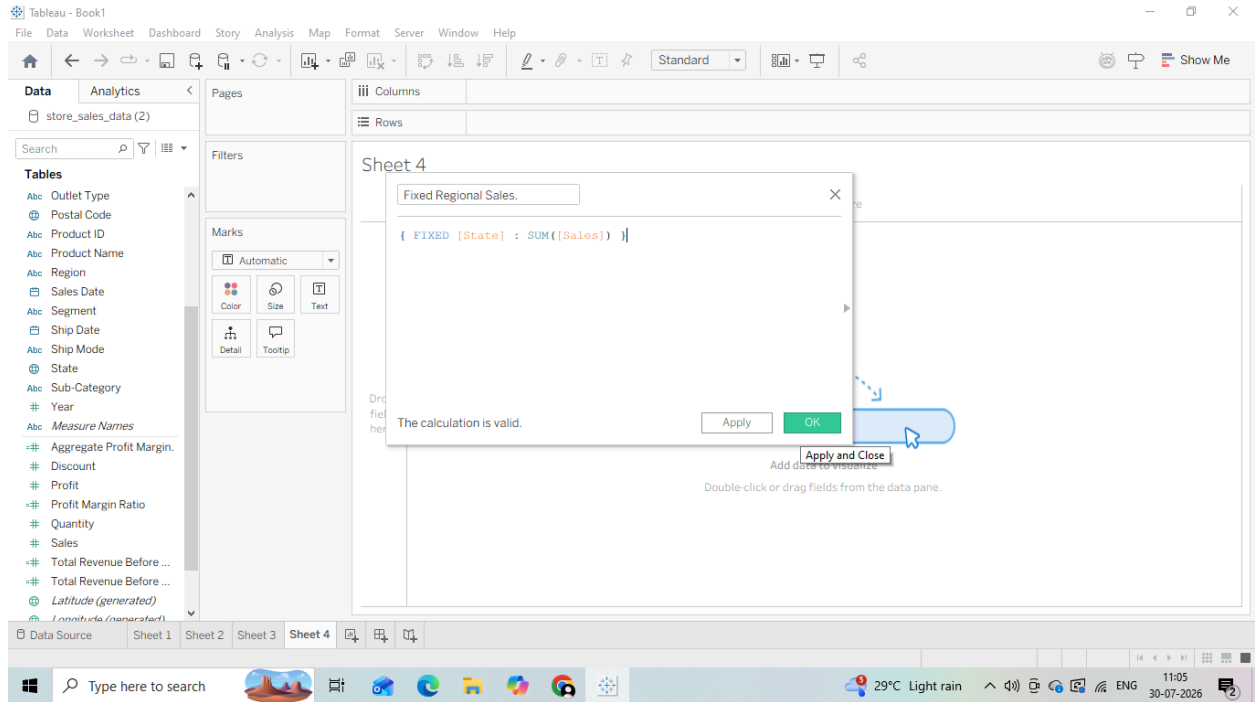
1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



3. In the calculation formula window, type the following expression using curly braces

`{: { FIXED [State] : SUM([Sales]) }` 4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.

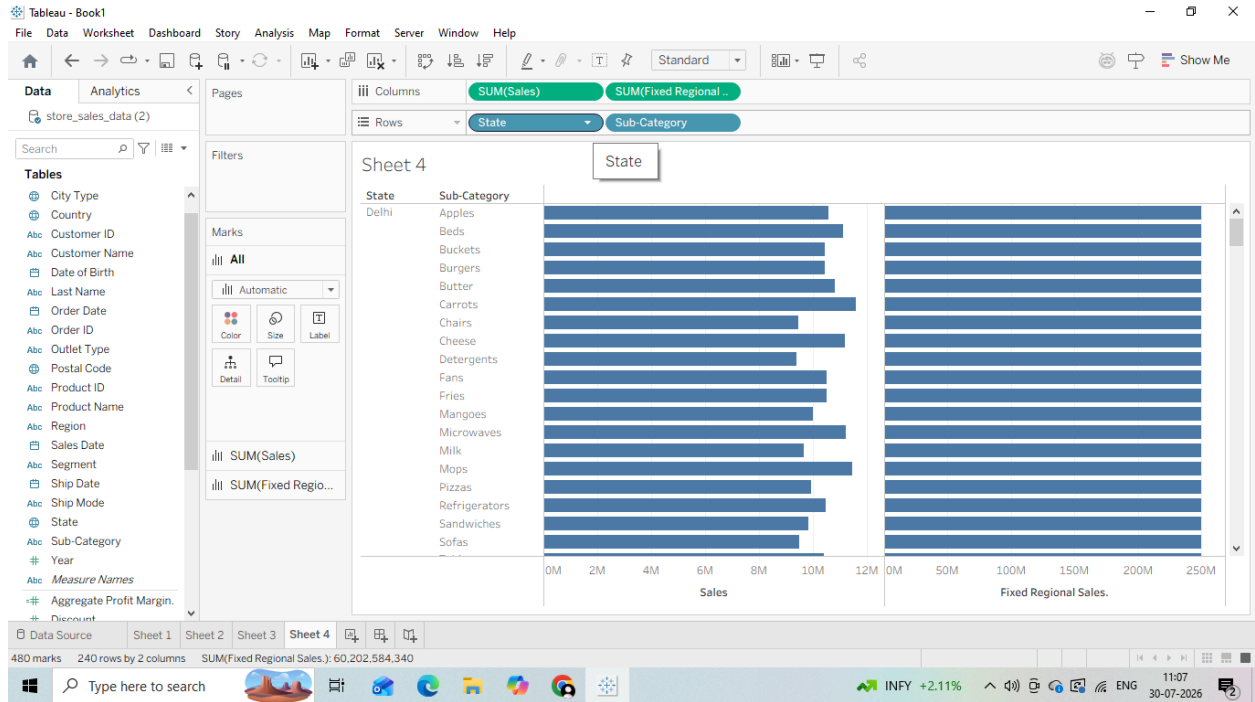
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2. Putting the Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State.
3. Drag the raw Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.

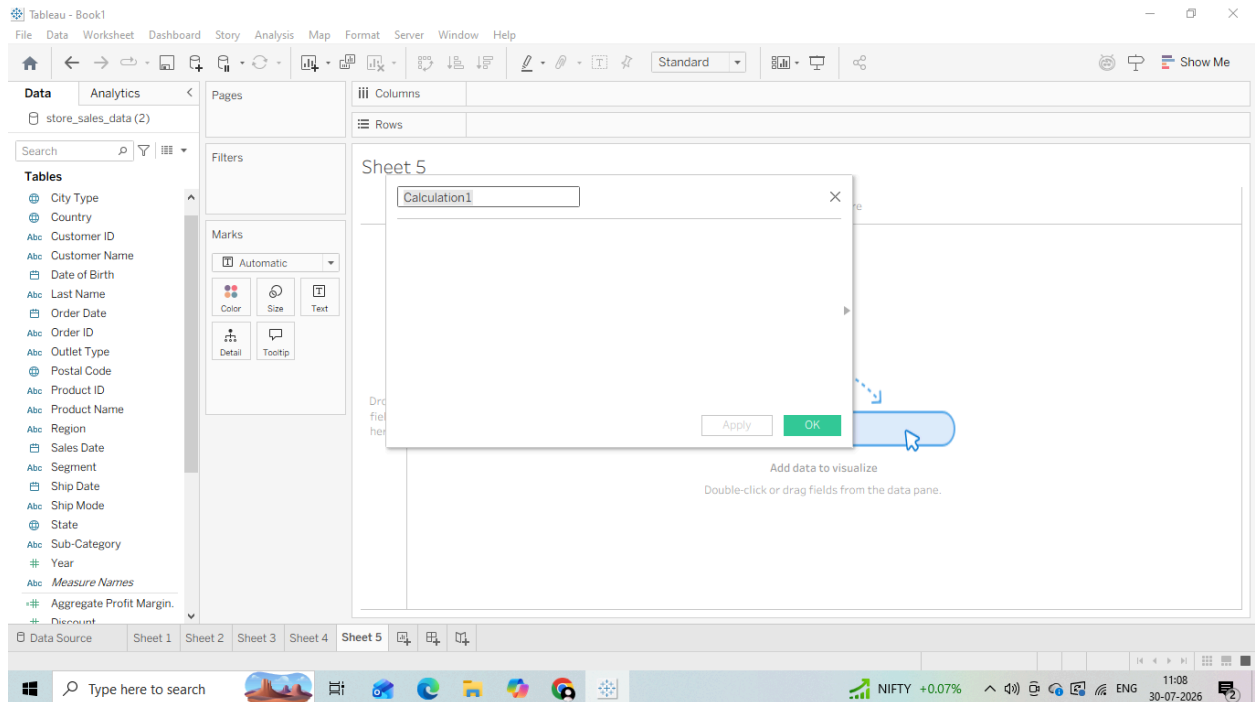
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E: Using INCLUDE and EXCLUDE LOD Expressions

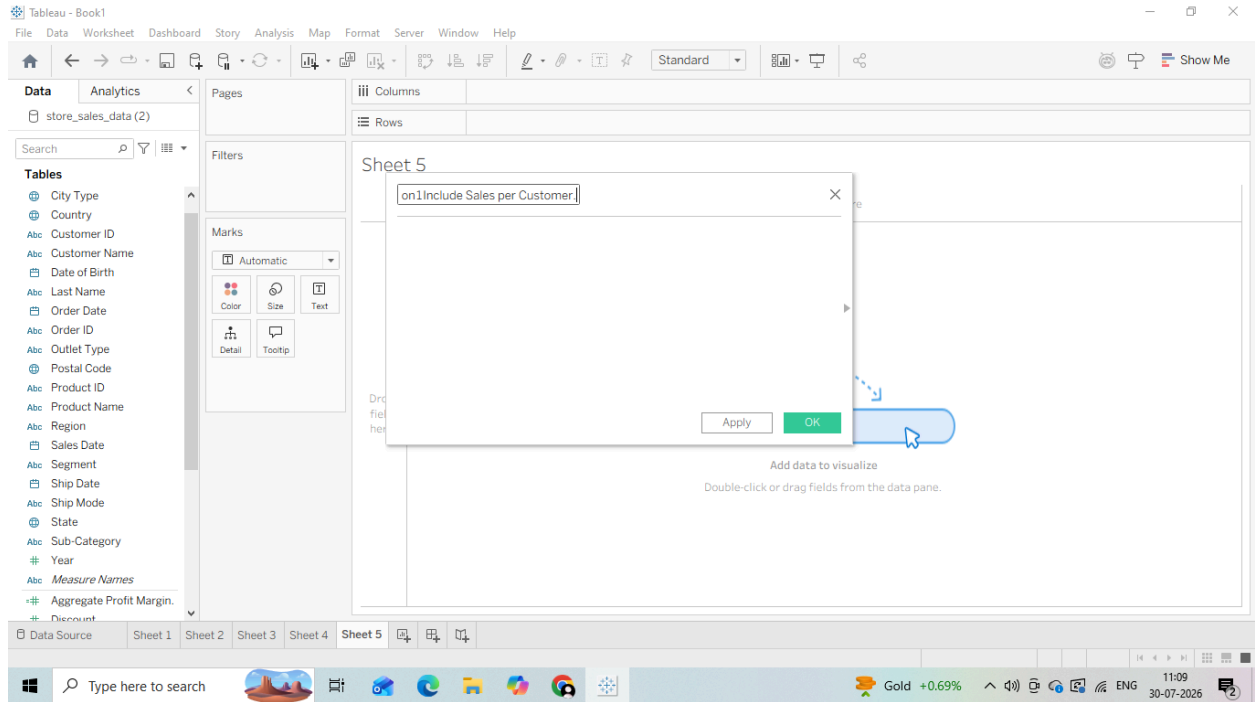
1. Creating an INCLUDE LOD Calculated Field

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...



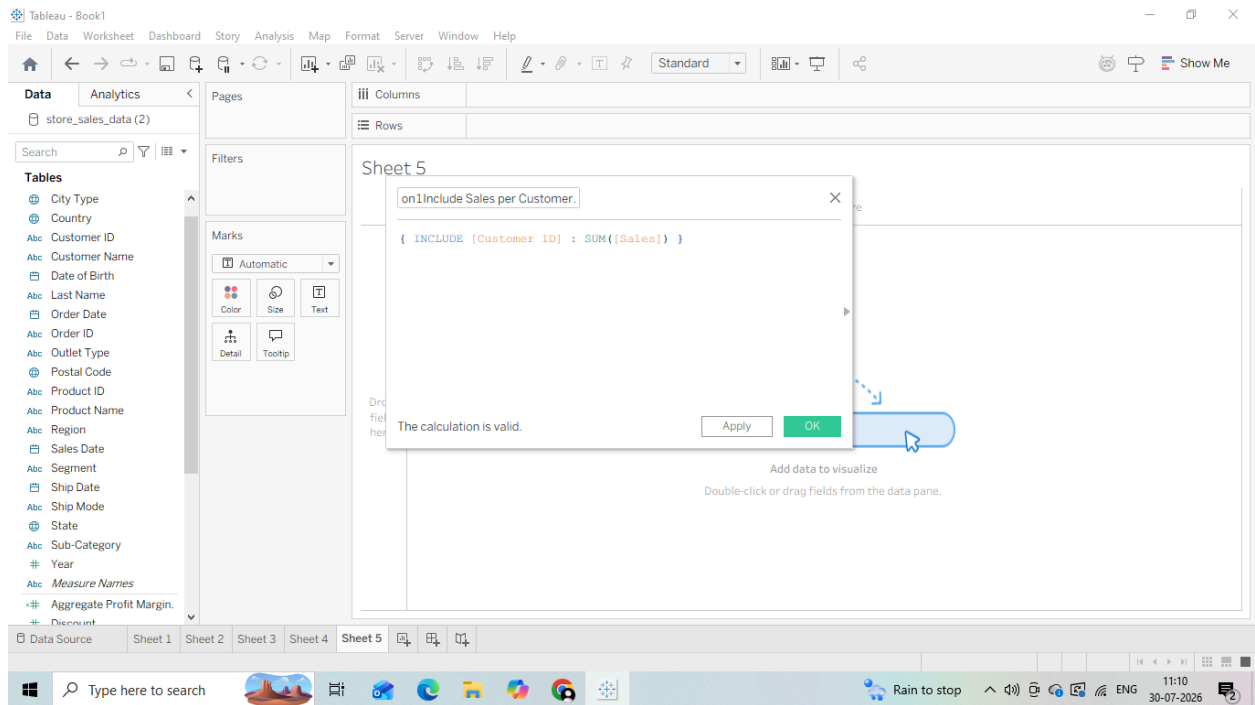
2. Name the field: Include Sales per Customer.

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3. In the calculation formula canvas area, input the following expression:

`{ INCLUDE [Customer ID] : SUM([Sales]) }`



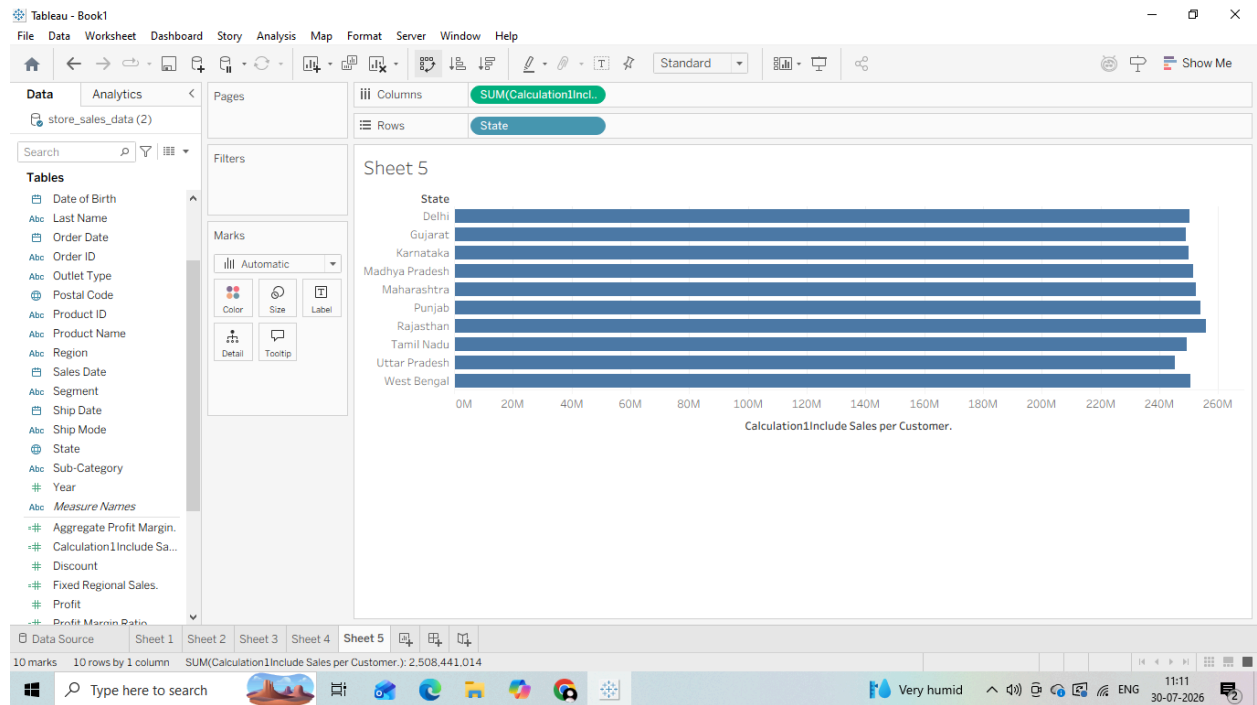
4. Click OK. 2. Putting the INCLUDE Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.

2. Drag your newly created Include Sales per Customer calculated field and drop it onto

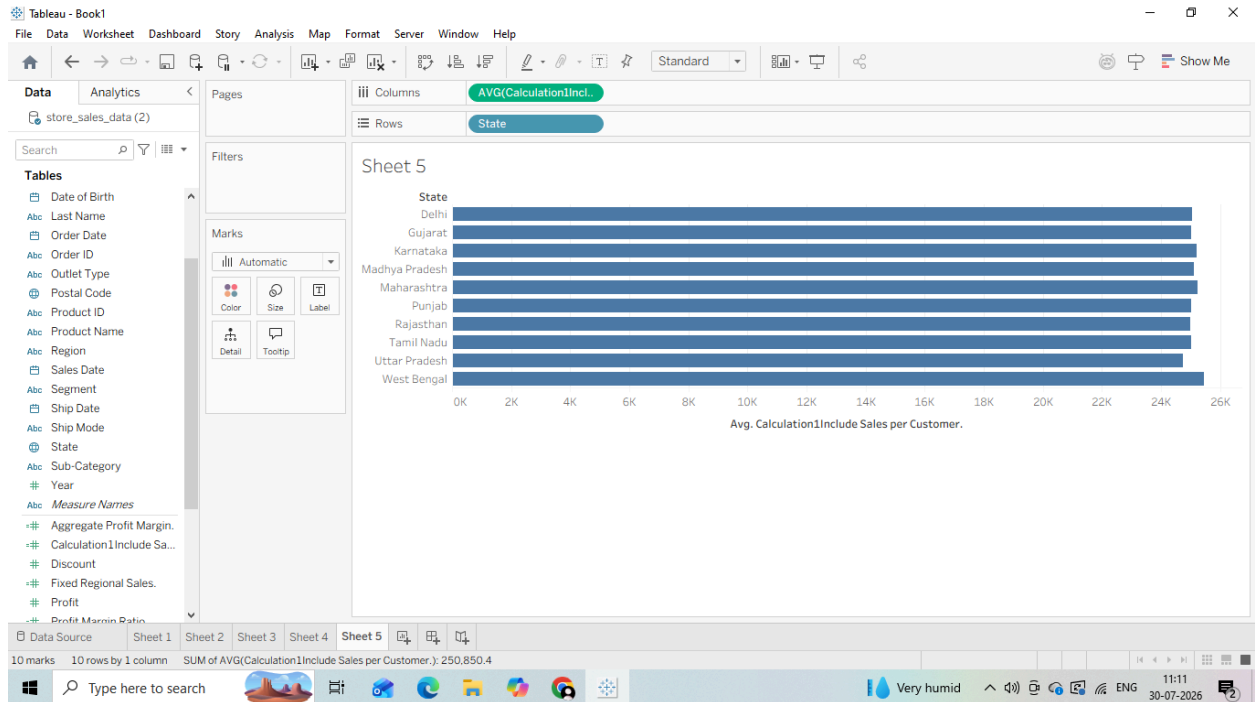
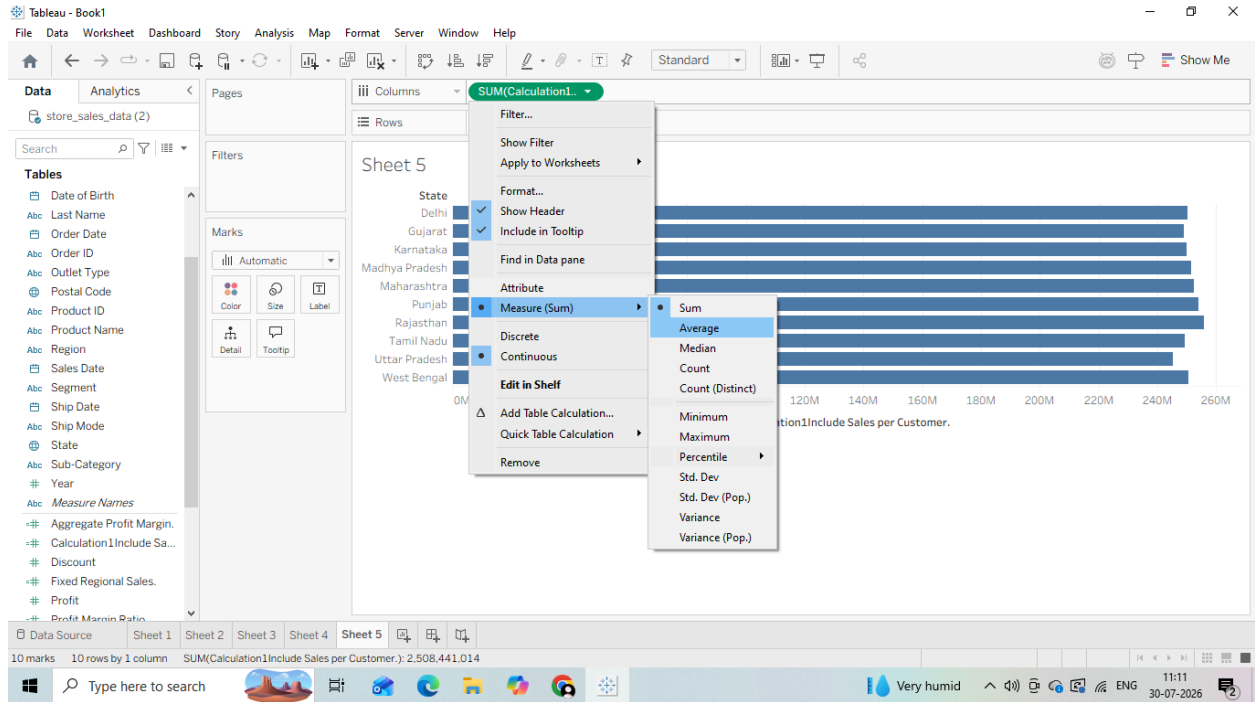
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the Columns shelf.



3. Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.

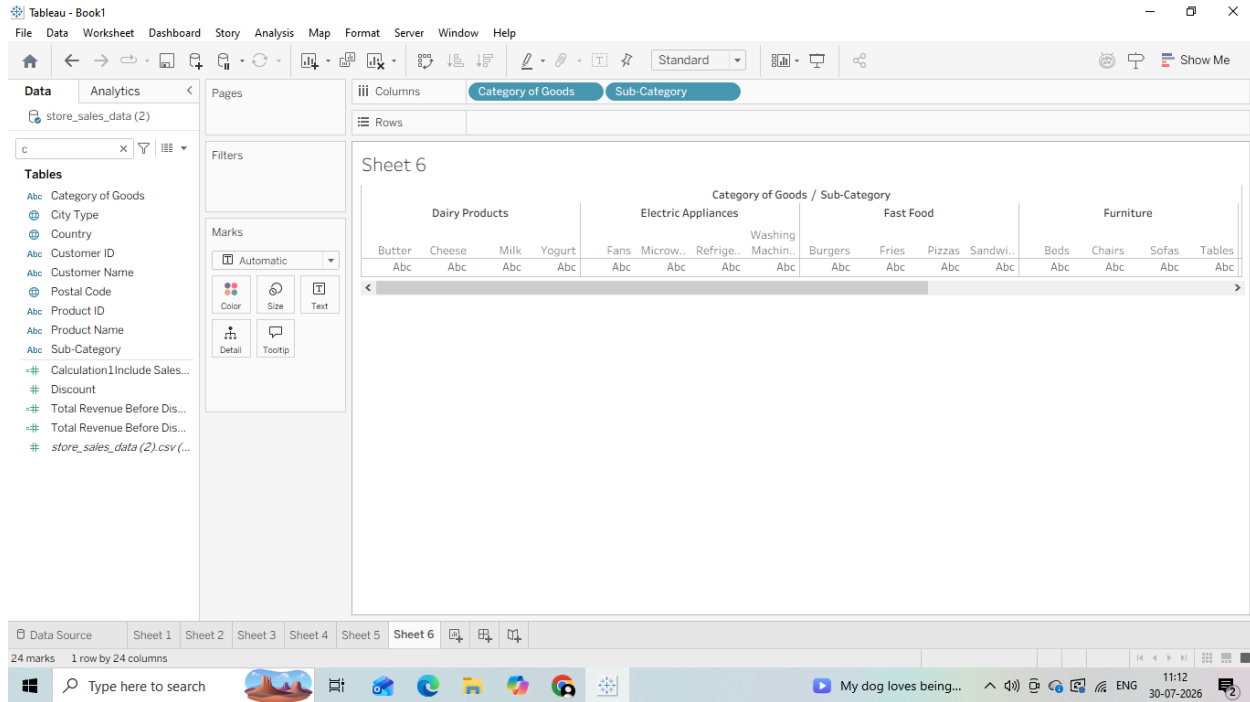
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3. Creating an EXCLUDE LOD Calculated Field

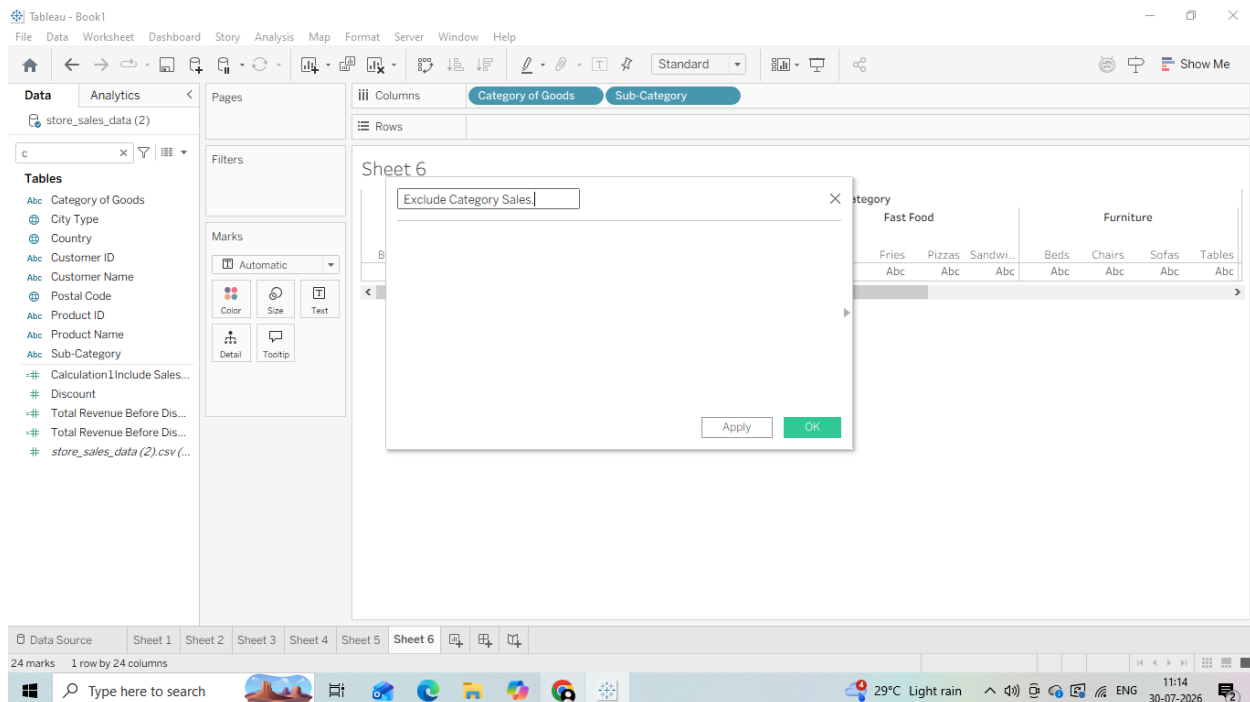
1. Open a new blank worksheet.
2. Drag Category of Goods onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).

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4. Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...

5. Name the field: Exclude Category Sales.



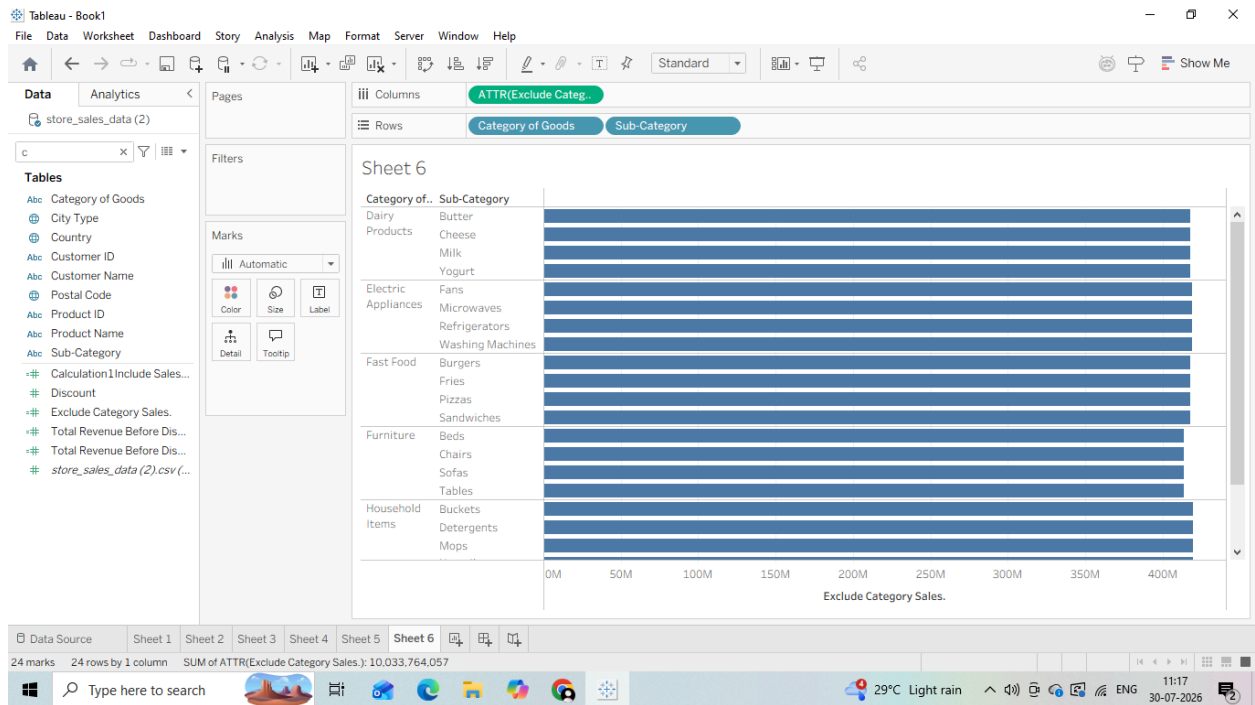
6. Input the following expression:

```
{ EXCLUDE [Sub-Category] : SUM([Sales]) }
```

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The screenshot shows the Tableau Desktop interface. The 'Columns' shelf contains 'Category of Goods' and 'Sub-Category'. The 'Rows' shelf is empty. The 'Marks' shelf is set to 'Automatic'. A dialog box titled 'Exclude Category Sales.' is open, displaying the calculated field formula: `{ EXCLUDE [Sub-Category] : SUM([Sales]) }`. The dialog box also indicates 'The calculation is valid.' and has 'Apply' and 'OK' buttons. The background view shows a table with columns for 'Category' (Fast Food, Furniture) and 'Sub-Category' (Fries, Pizzas, Sandwiches, Beds, Chairs, Sofas, Tables).

7. Click OK and drag this new field onto the Columns shelf.v



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F: Creating and Using Parameters for Dynamic Analysis

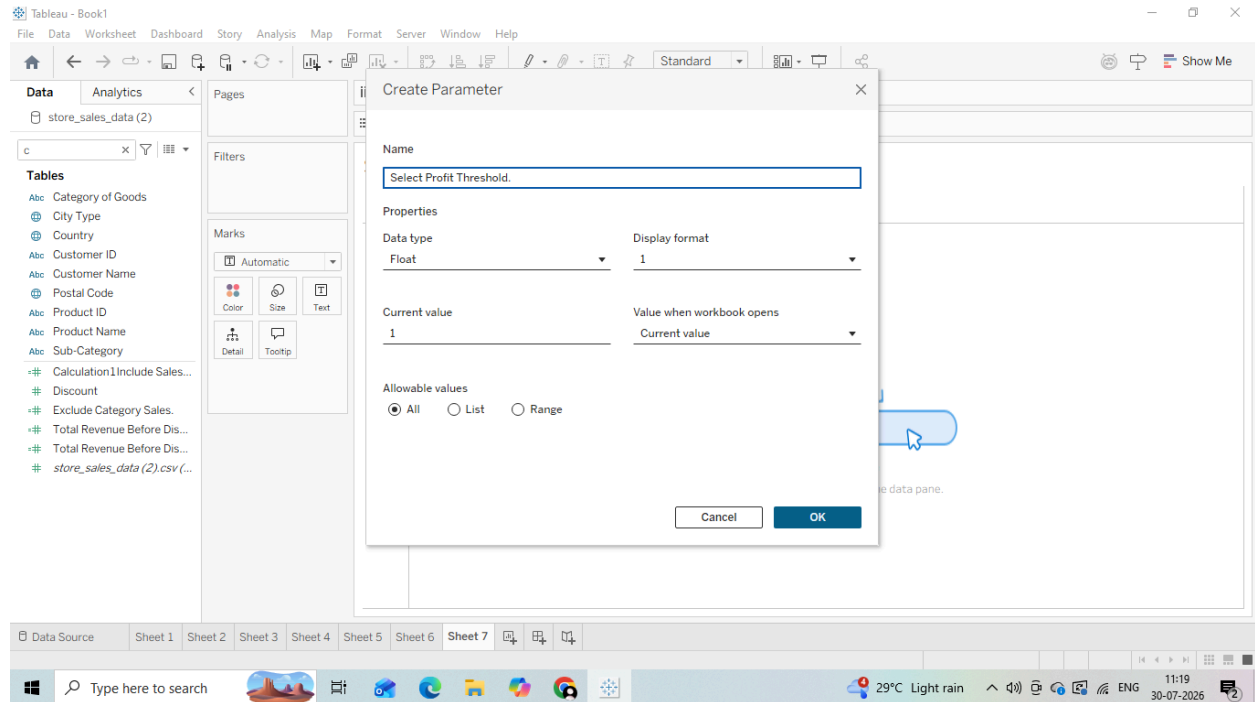
1. Creating a Parameter

Unlike calculated fields which evaluate data statically, a Parameter is a dynamic variable or workbook input control block (like a slider or dropdown) that allows users to alter calculations or values on the fly without changing the underlying architecture.

Let's build a parameter that lets users set their own dynamic Profit Target threshold:

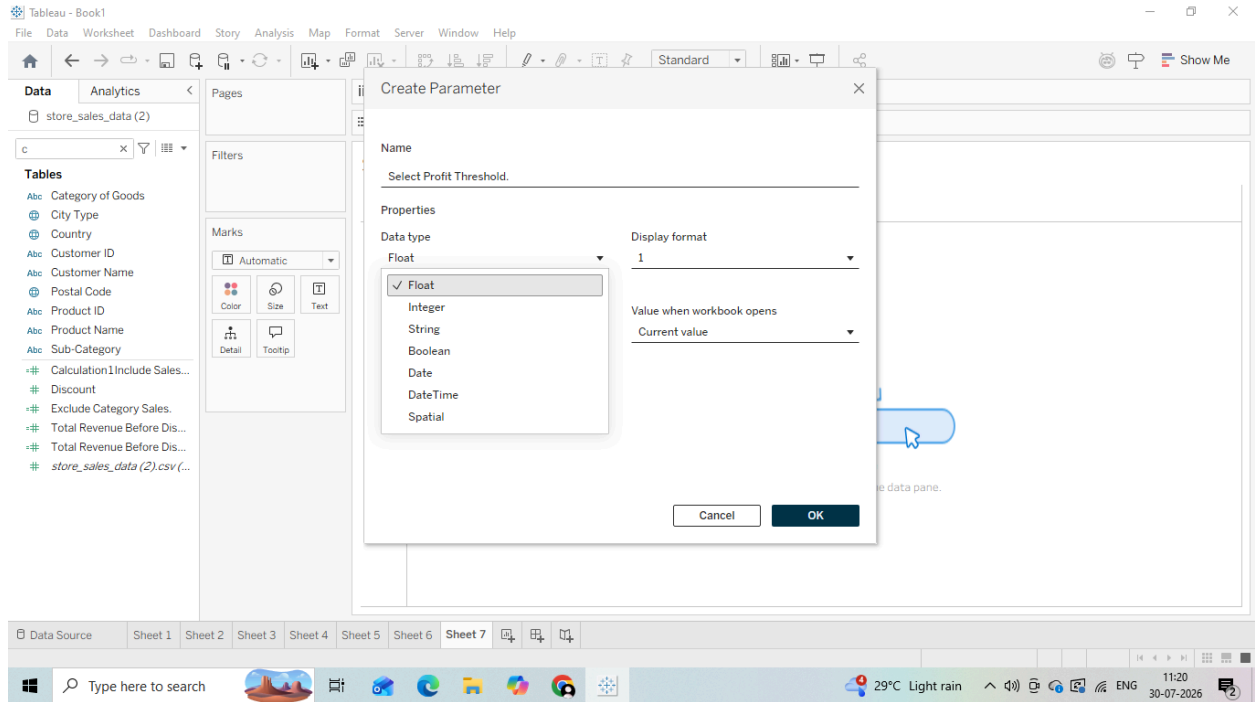
1. Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose Create Parameter.

2. Name the parameter: Select Profit Threshold.

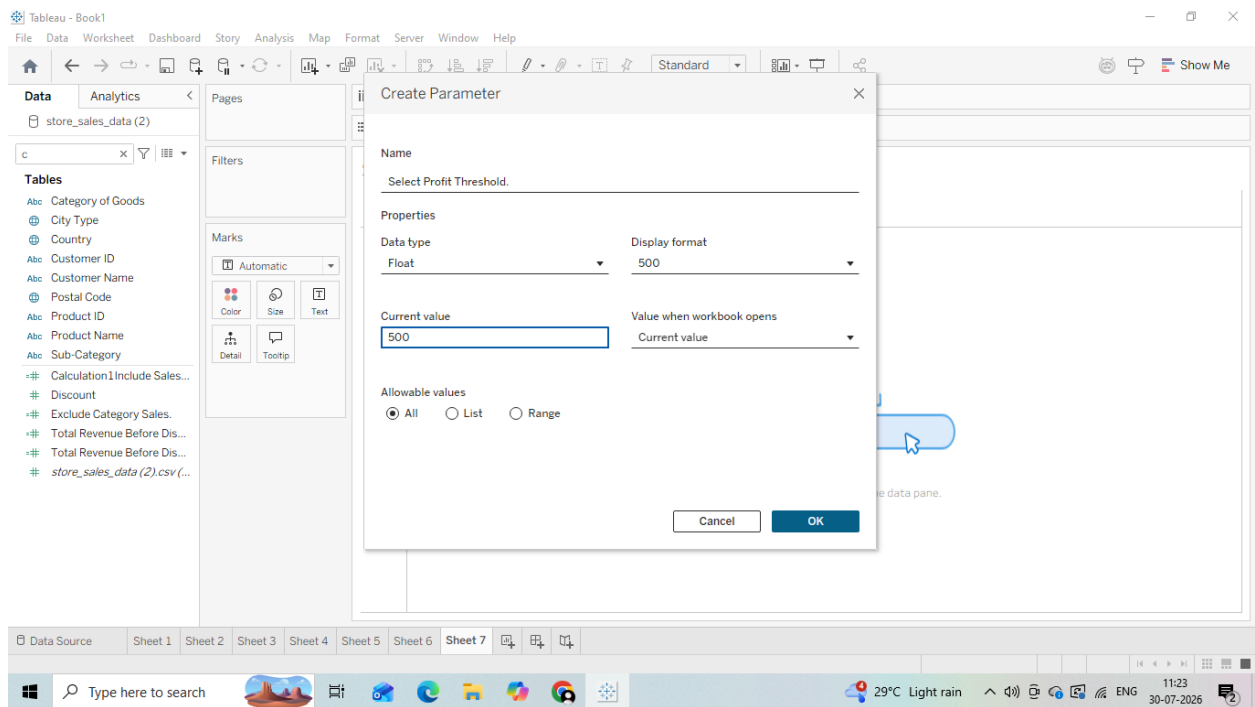


3. Set the Data Type to Float (or Integer).

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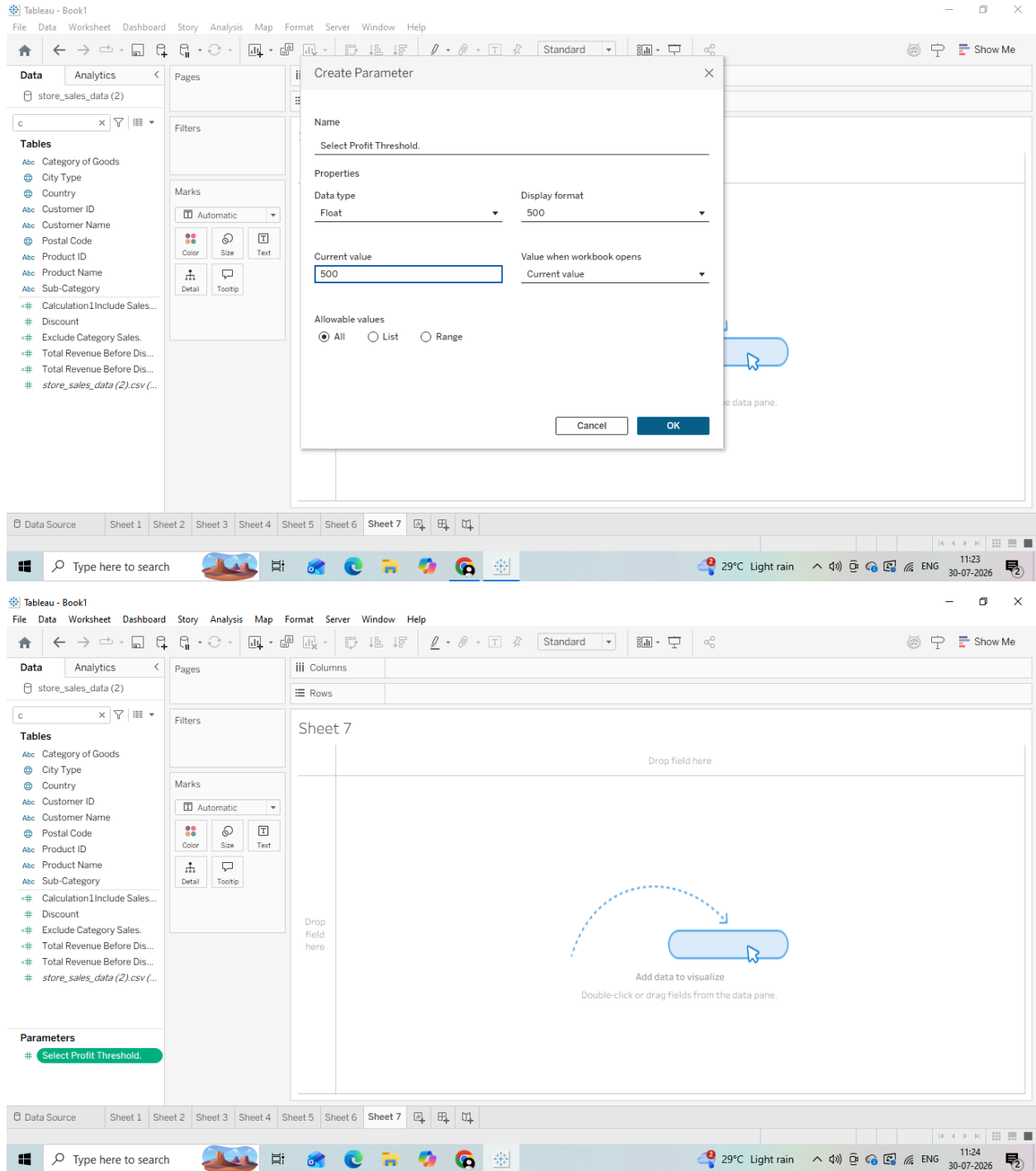


4. Set the Current Value to 500.



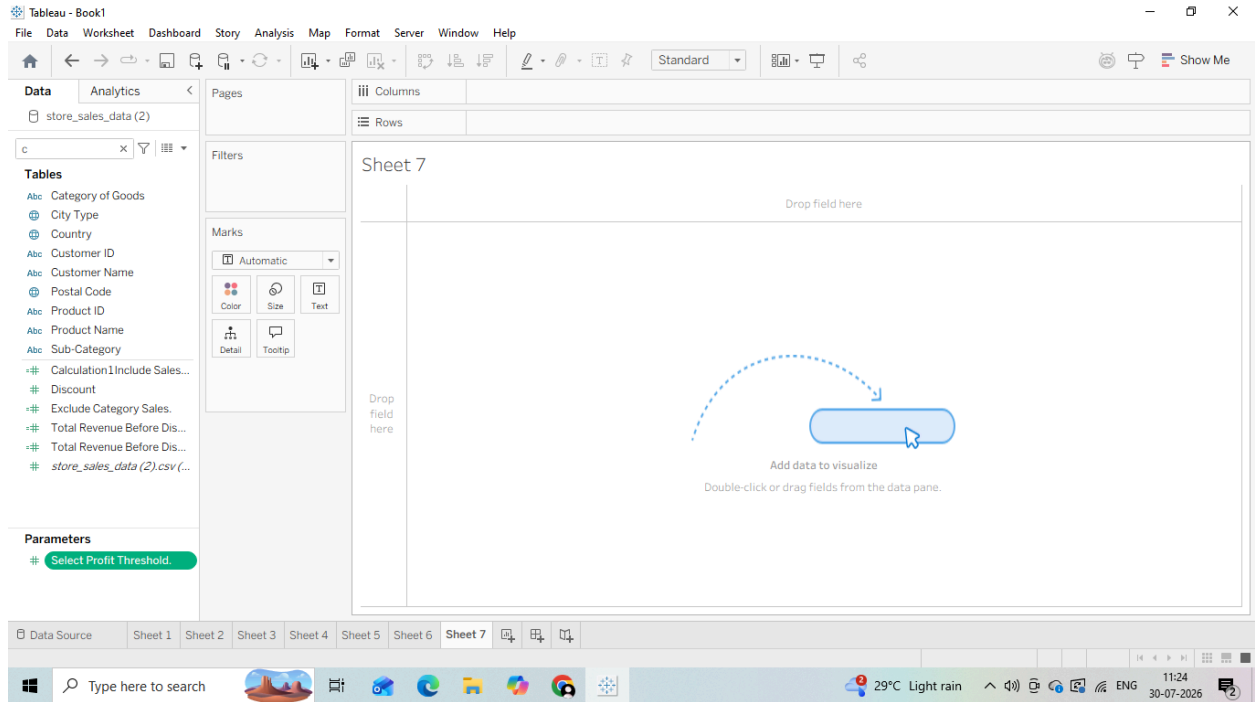
5. Keep the Allowable values set to All and click OK.

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6. Notice a new section titled Parameters appears at the bottom of your Data Pane, showcasing your new selector.

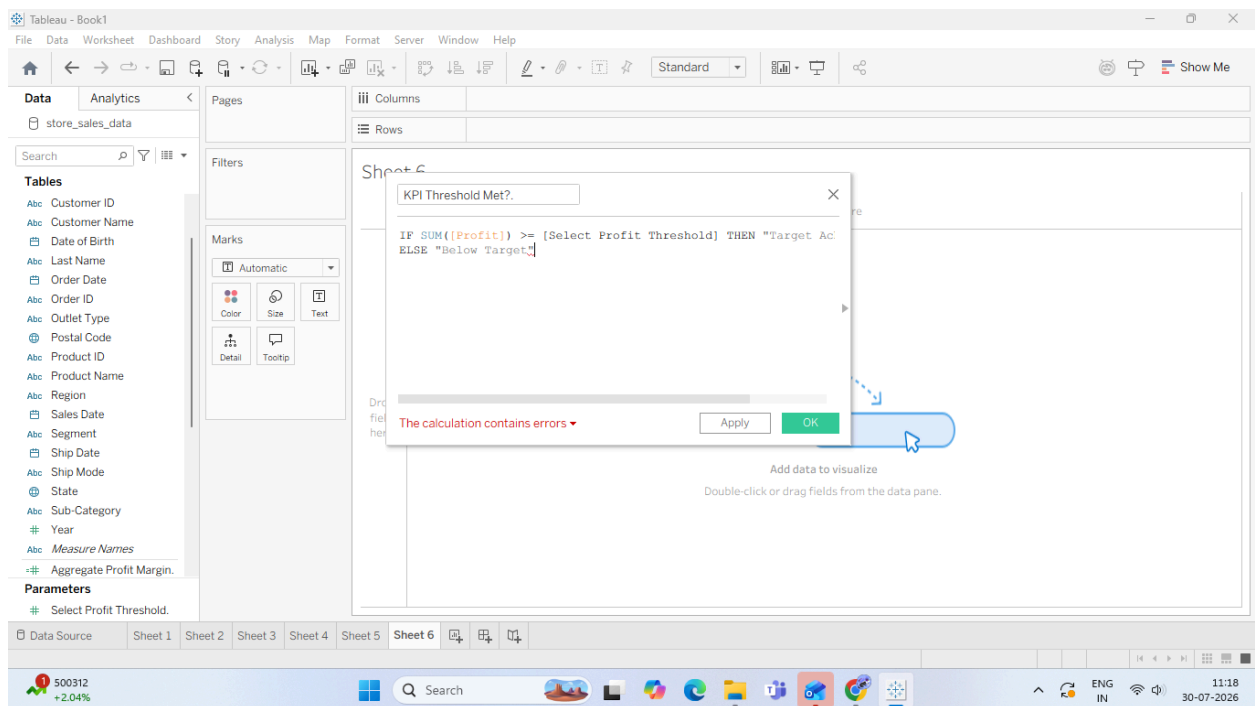
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2. Linking the Parameter to a Calculated Field

A parameter does absolutely nothing until you actively tie it to a field layout or formula. Let's create a conditional formula that flags categories based on the parameter's current value:

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field...



2. Name this field: KPI Threshold Met?.

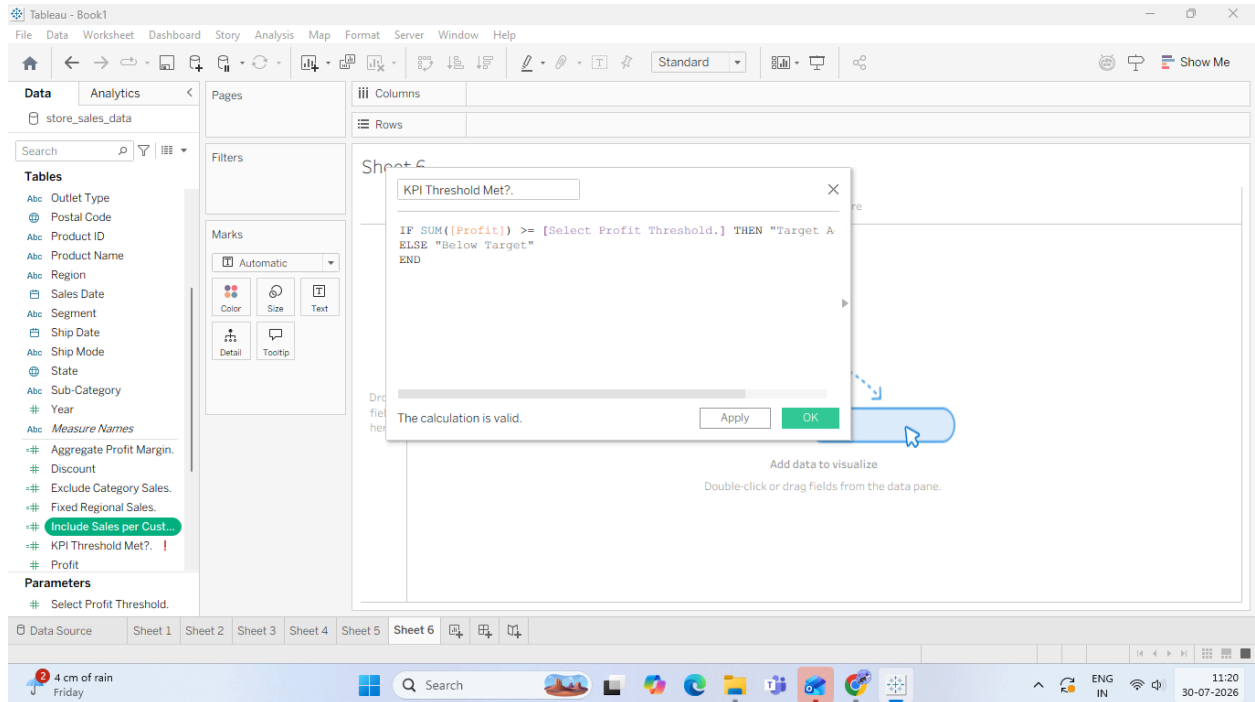
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3. Input the following logical IF-THEN conditional statement:

IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved"

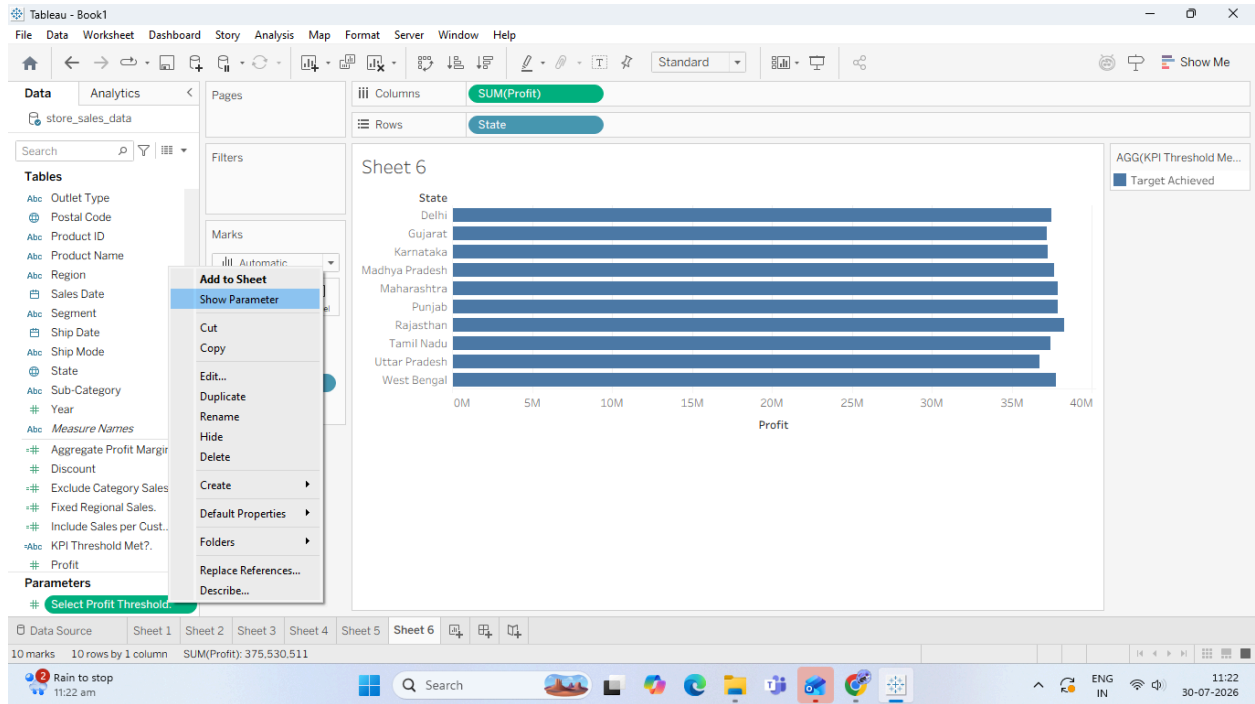
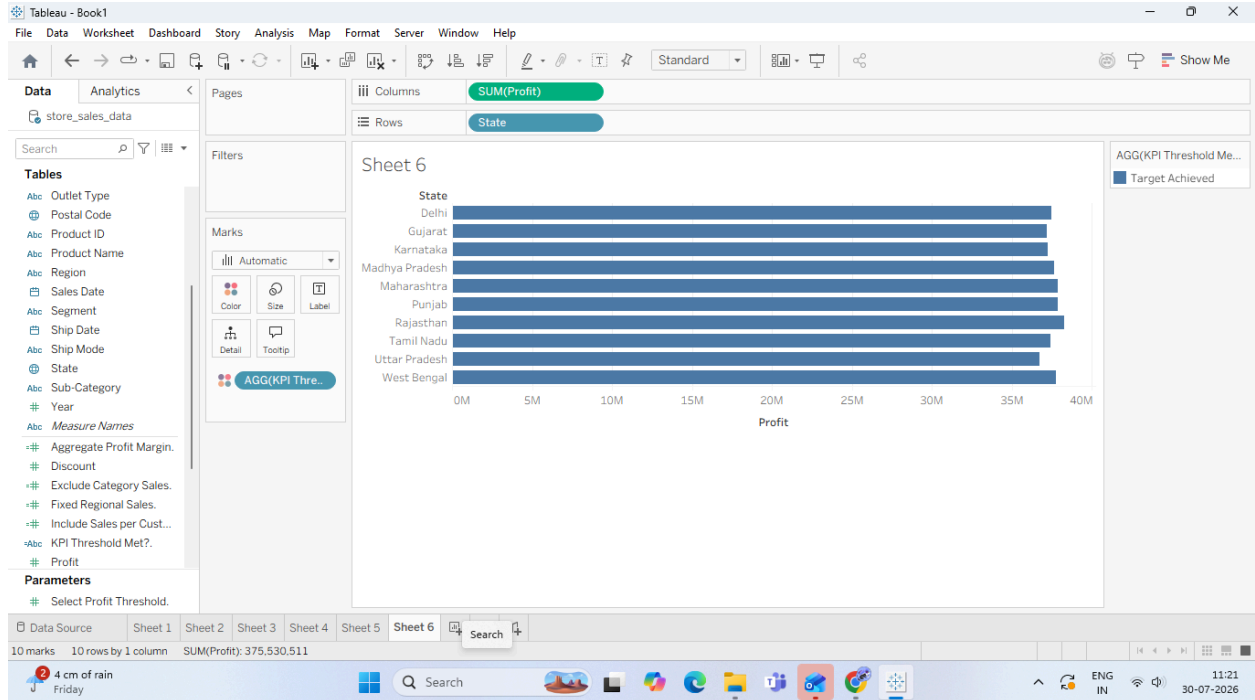
ELSE "Below Target"

END (NOTE: THE SELECT PROFIT THRESHOLD SHOULD TURN INTO PURPLE IF IT IS NOT TURNING IN PURPLE PLEASE RETYPE IT AGAIN)

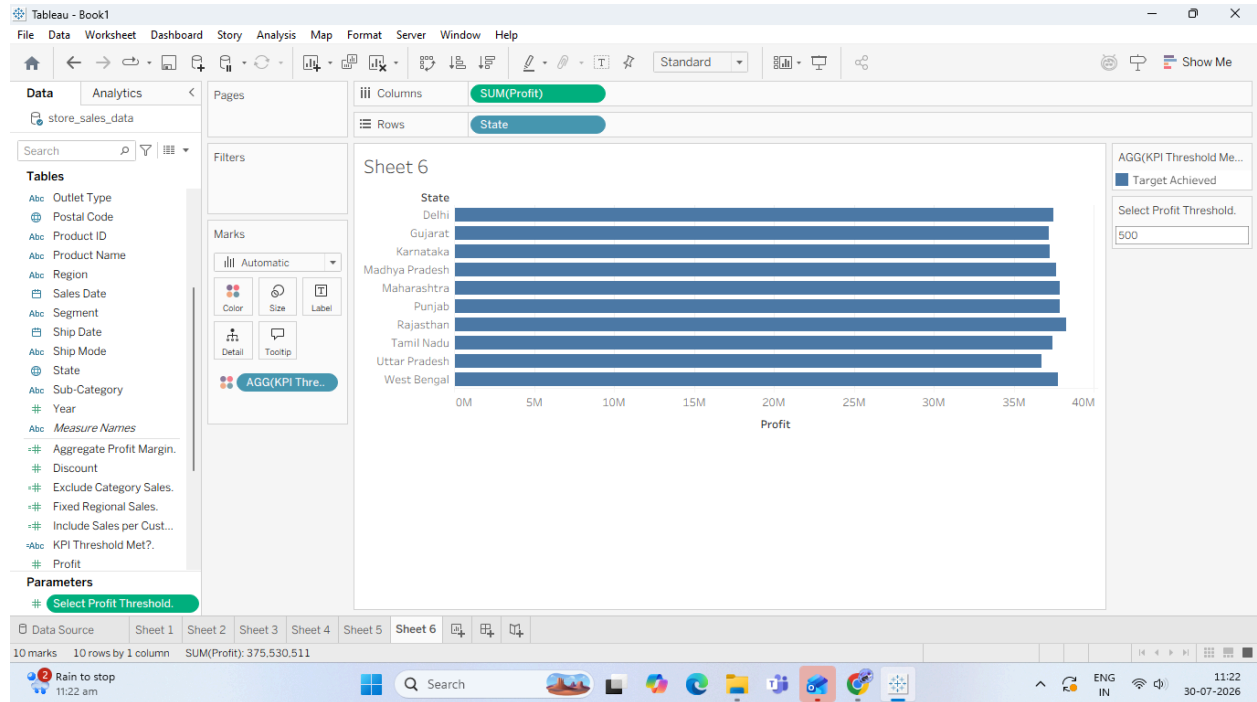


5. Show the Parameter Control: Go down to the very bottom of the Data Pane under the Parameters header section. Right-click on your Select Profit Threshold parameter block and choose Show Parameter Control.

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G: Apply Calculations to Fix Data Issues and Enhance Insights

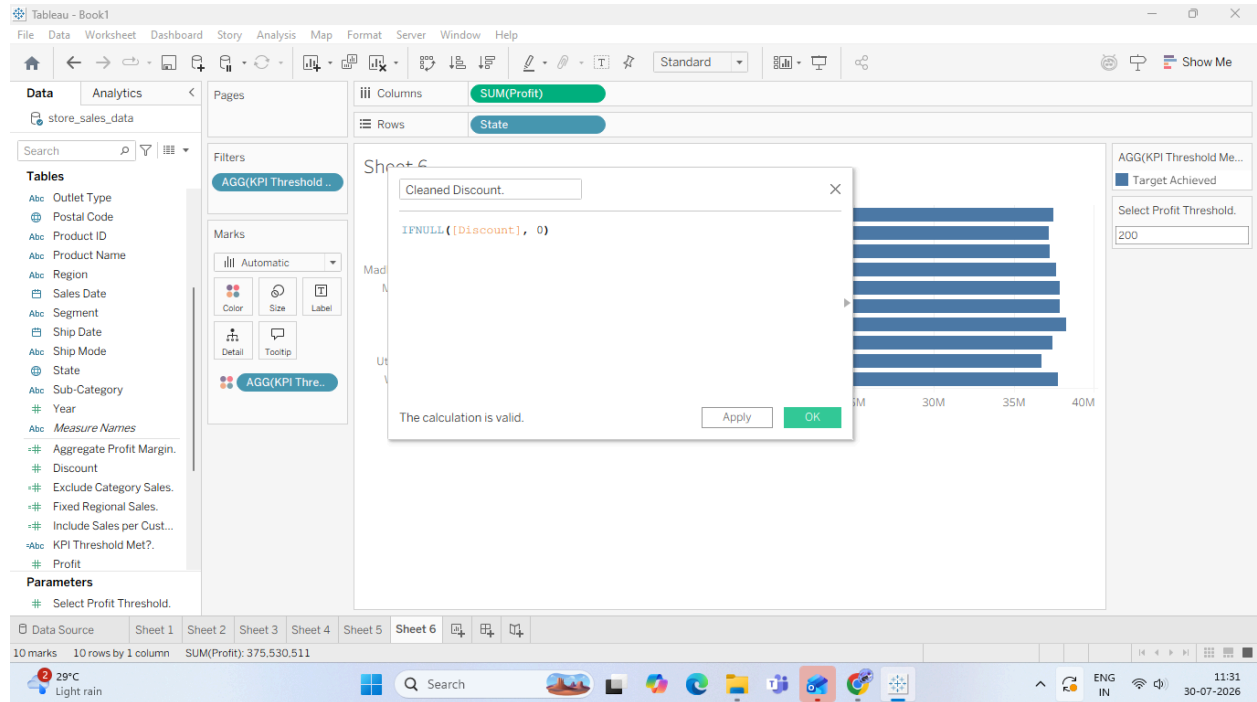
1. Handling Null or Missing Values with IFNULL

In enterprise datasets, missing data or null entries can cause calculations to break or display empty spaces in charts. We can use a logical calculation to substitute a clean value whenever a null occurs.

Let's clean up potential missing indicators in our data:

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...

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2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

The screenshot shows the Tableau Desktop interface. On the left, the 'Data' pane lists fields under 'Tables' (Outlet Type, Postal Code, Product ID, Product Name, Region, Sales Date, Segment, Ship Date, Ship Mode, State, Sub-Category, Year) and 'Parameters' (Select Profit Threshold). The 'Columns' shelf is empty, and the 'Rows' shelf contains 'SUM(Sales)'. The 'Marks' card is set to 'Automatic'. A dialog box titled 'Standardized city type.' is open, displaying the calculation `UPPER([City Type])`. The dialog shows a message 'The calculation is valid.' and 'OK' and 'Apply' buttons. The background view is partially obscured by the dialog box.

